

Practitioner's Guide to Group Testing

The definitive guide for practitioners implementing group testing into practice.



Some Words from the Experts



Cullen McGowan
Managing Director
Prep Baseball

[in](#) [X](#) [@cullen_mcgowan](#)

“

Scalable testing allows us to evaluate players on a level playing field. Regardless of geography or prior exposure, every athlete deserves access to the same objective standards and credible feedback. That belief has driven us to intentionally build a broad, diverse geographic footprint so athletes don't have to chase exposure.

By running more than **1,000 showcases annually across 45 states – and testing over 50,000 players each year** – we make it easy for athletes to access Prep Baseball events close to home while still competing against a national benchmark. To support that scale, **we rely on technology that's durable, resilient and consistent, allowing us to efficiently reach players regardless of the facility or infrastructure in their backyard.**

Operating at a national scale enables us to combine technology, standardized testing protocols and on-the-ground expertise to capture one of the most comprehensive datasets in amateur baseball. Few organizations have the infrastructure to test high school players this consistently, and that scale is what allows Prep Baseball to turn raw performance into meaningful feedback and insights in our performance tier product.”



Dr. Samuel Spinelli
High Performance Coordinator,
Thrive Kelowna

[in](#) [@](#) [X](#) [@samspinelli.dpt](#)

“

Testing is only as good as the data it produces. When testing large groups, the manual hurdles of the past often led to a lack of objective guidance, leaving practitioners to rely on intuition rather than high-fidelity evidence. With VALD, we've eliminated that compromise.

Our group combine process eliminates these friction points, leveraging automated technology to streamline the athlete experience. This allows our staff to focus on athlete communication and test execution rather than manual data entry. Our resulting metrics are clean and organized, enabling us to provide immediate, actionable feedback to help identify at-risk populations and guide specific training interventions.

By applying VALD's technology to our combine methodology, we have redefined our facility's capacity to deliver elite-level insights at scale.”



Ronny Torres
VP of Brand Growth and Development,
Student Sports

[in](#) [X](#) [@RonnyTorres_](#)

“ We’ve tested thousands of elite high school athletes across hundreds of events dating back decades. However, **since introducing VALD technology, including SmartSpeed, ForceDecks and DynaMo, we’ve been able to take our collection and evaluation processes to the next level.** ”

Today’s amateur athletes are stronger, more dynamic and better prepared than ever. Providing them with objective data and instant feedback has added another valuable layer to their experience – something that has always been a priority for us. The best athletes are eager to compete and know where they stand, and VALD technology has not only optimized high-volume testing efficiency for us but, more importantly, has given athletes the accurate information they need to continue elevating their game. ”



Taylor Evernden
BSc Kin., MSc, Founder, Director of Performance,
ORKA Performance

[in](#) [@](#) [X](#) [@yycstrengthcoach](#)

“ In group and combine testing environments, the value of performance technology is in its ability to scale. **When you’re assessing hundreds of athletes, efficiency and consistency matter more than novelty.** The right tools allow you to collect meaningful data quickly and actually apply it beyond the testing day. ”



Anna Cruse
Director of Applied Health and Performance Science
University of Utah Athletics

[in](#) [X](#) [@acruse_](#)

“ When baseline or combine testing for large groups of athletes, it’s imperative that the tests are repeatable, reliable, efficient and valid. Whether using the results to evaluate programming and interventions or to track progress over time, valid and reliable technology and assessments provide the confidence to know that you’re making the best decisions every time. ”

Efficient technologies make it easy for practitioners to over-test. **Choose only impactful tests that make sense for your group or sport, and ensure that the data collected will actually be used.** When planning a group testing day, remember that the information yielded from testing only has value if it’s used to inform decision-making or influence change. ”

Purpose of This Guide

This guide is designed to help practitioners confidently plan, run and interpret testing sessions with large groups and organizations.

While many practitioners are familiar with single-participant testing workflows, managing large groups introduces unique logistical and technical challenges. These challenges can be overcome by applying the insights and strategies outlined below, ensuring that both data quality and participant experience remain high during testing sessions, even as testing volume increases.



This guide aims to make group testing more efficient, consistent and actionable. Whether testing ten participants in a training environment or hundreds at a talent identification event, practitioners will be confident in taking the steps to deliver high-quality testing experiences at scale.

This guide aims to make group testing more efficient, consistent and actionable. Whether testing ten participants...or hundreds...

The Role of Group Testing in Modern Practice

Across sport, health and education, practitioners are increasingly required to collect more data from more participants at a higher frequency. However, time, staffing and facility constraints often limit the opportunity for one-on-one assessments, making efficient group testing a sought-after skill for performance practitioners.

This need is driven by an opportunity to create participant buy-in through meaningful feedback and provide longitudinal data that supports better decision-making. Ultimately, these outcomes not only enhance participant outcomes but also improve an organization's return on investment (ROI) for measurement technology.

Ultimately, [group testing events]...**enhance participant management [and] improve an organization's ROI** for measurement technology.



Modern technologies, such as ForceDecks, SmartSpeed and other VALD systems, make this process more achievable than ever. These systems can streamline data collection and analysis, allowing practitioners to capture meaningful insights from large numbers of participants without sacrificing quality or accuracy.

Who Is This Guide for?

This guide is for anyone responsible for planning or providing group testing sessions, such as:

- Strength and conditioning coaches
- Sport scientists and performance analysts
- Rehabilitation professionals
- Workplace health and safety professionals
- Teachers and academy coordinators
- Academic researchers
- Talent identification and recruiting staff

Each section is designed to be adaptable – equally relevant to high-performance sport, community health programs or academic settings.



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In this document, we will **cover a wide range of topics and answer some common questions**, including:

Click any heading to jump to section

Testing Order

Typical Group Testing **Environments**

Event Flow and Station Management

Selecting the **Right Tests**

Selecting the **Right Technology**

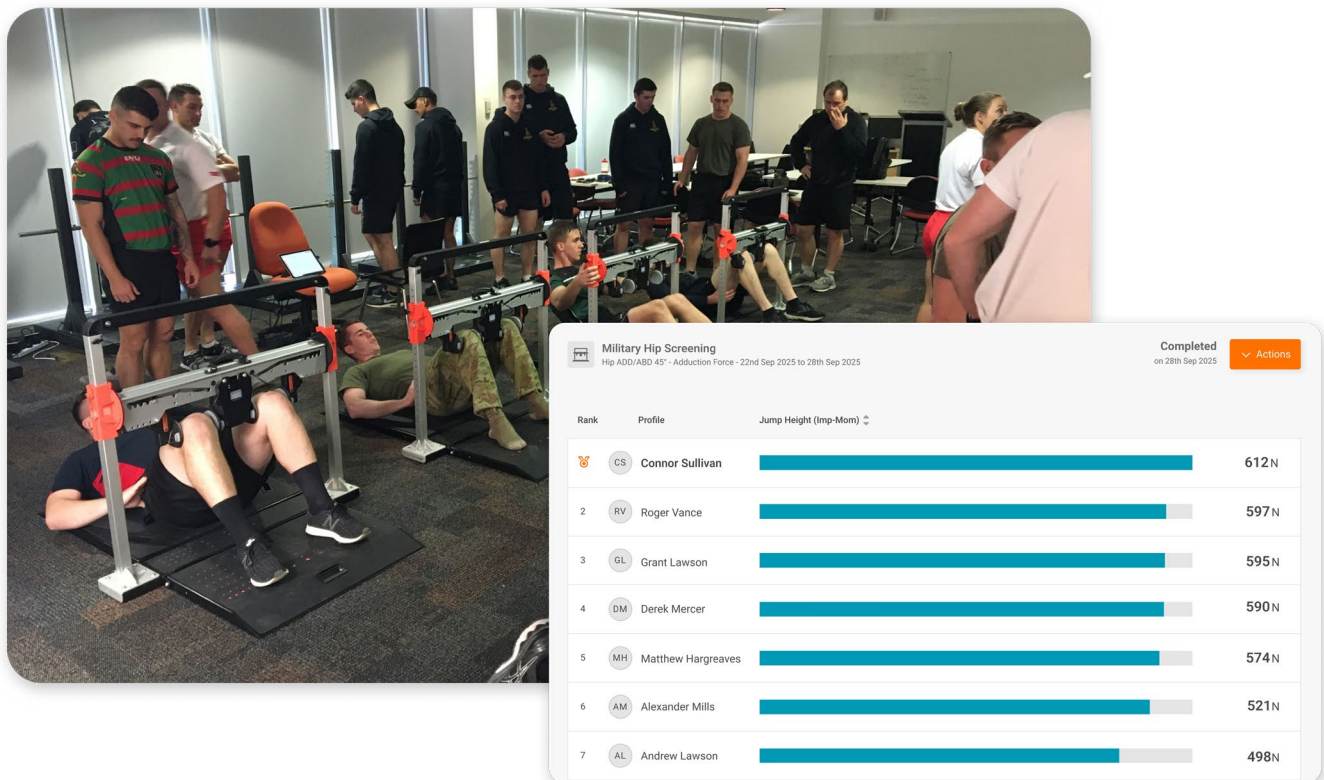
Choosing the **Right Reporting Path**

Group Testing Fundamentals

This section introduces the core principles of group testing, outlining where it is most commonly applied, how technology enables scale and what key considerations are involved in implementing measurement strategies across large groups.

What Is Group Testing?

Group testing is the process of assessing multiple participants within a single session using standardized protocols, equipment and a clearly defined testing flow. It allows practitioners to collect objective data efficiently in environments where individual testing is impractical, such as teams, workplaces or large classes.



Academic environment testing hip strength while using [VALD Hub's Leaderboard](#) function to encourage engagement.

[Group testing] allows practitioners to **collect objective data efficiently in environments where individual testing is impractical...**

When supported by appropriate preparation and scalable measurement technology, group testing enables high-volume data collection while maintaining data quality across health and performance environments.

Typical Group Testing **Environments**

Group testing is applied across a range of settings, including sport, healthcare, workplace wellness and education. While specific tests and metrics may vary depending on the population, measurement technologies like [ForceDecks](#), [ForceFrame](#) and the suite of [VALD systems](#) provide individual and group testing solutions across a wide range of participant capabilities.



Sports Teams

Professional, collegiate and amateur sports organizations regularly rely on group testing for preseason profiling, in-season monitoring and end-of-season benchmarking. For example, a professional team may complete weekly [countermovement jump \(CMJ\)](#) testing on ForceDecks before training. Thirty players can be tested in just over 20 minutes, providing consistent data to track neuromuscular readiness and training response.

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Schools and Academies

In education or talent-development settings, time constraints and large numbers are constant challenges. Up to 200 participants can be assessed over a single day or across multiple days for qualities such as lower-body power or sprint performance.

Tools like [SmartSpeed Plus](#) and [SmartJump](#) enable practitioners to quickly assess sprint and jump performance across all participants, while easily ranking and comparing them with analysis solutions like [Group Dashboards](#) and [Norms](#) available in [VALD Hub](#).

Combines and Talent Identification Camps

Large-scale events such as draft combines or regional screening days require standardized, repeatable testing across large populations. Testing may occur on a single day, over multiple days or concurrently across various locations.

Measurement technologies, including dynamometry systems like [NordBord](#), support this process by enabling portable, validated and repeatable data collection. These services may be delivered internally by team or league staff, such as at the [NBA Draft Combine](#), or externally by private organizations, such as the [Prep Baseball Super 60](#) event.

Measurement technologies...support this process by **enabling portable, validated and repeatable data collection.**

Workplace Wellness

Beyond performance sport, group testing plays a key role in workplace wellness programs and labor-related injury screening. Objective measurement technologies allow organizations to efficiently assess strength, movement and physical capacity across large employee groups.

For example, markerless motion capture systems such as [HumanTrak](#) can be used during on-site screening days to assess movement quality in a few seconds, providing immediate, objective feedback to participants and employment stakeholders.

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Benefits of **Technology-Enabled** Group Testing

Objective measurement technology allows group testing to scale without compromising data quality by streamlining setup, automating data capture and reducing manual processing. These capabilities help practitioners maintain consistent testing standards when working under time constraints. The following list outlines key technology-driven benefits that support effective group testing:



Standardization

Using consistent equipment, protocols and coaching cues ensures data comparability. Validated systems, such as VALD's, allow results to be directly compared across participants, sessions and even organizations.



Benchmarking and Comparison

Aggregating group data in real time enables practitioners to benchmark participants against team averages, positional Norms and larger normative datasets. This supports both individualized feedback and strategic decision-making.



Engagement and Motivation

Testing in groups naturally drives competitiveness. Systems that provide instant, objective feedback that participants can see immediately often enhance effort and buy-in. Displaying live leaderboards or team averages can further increase engagement.



Resource Management

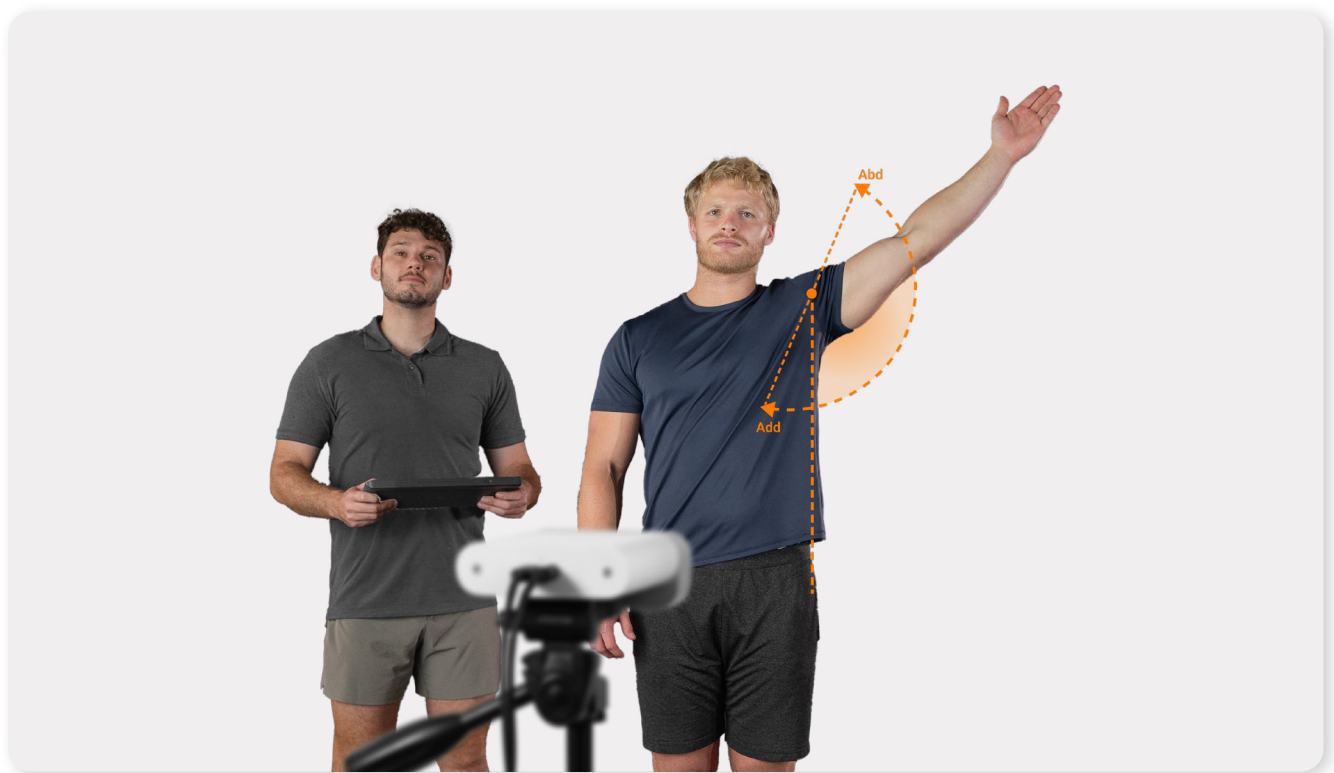
Automating data logging reduces manual entry and human error. Practitioners can spend less time recording and more time coaching, observing and interacting with participants.

For more information on improving the efficiency and standardization of group testing processes and systems, see the [Additional Resources](#) section at the end of this guide.

Measurement Technologies for Group Testing

Well-designed testing technology supports consistent execution across group testing environments, from small academies to full rosters and combine-style testing events. By reducing setup time and automating data capture, these systems allow practitioners to focus on quality assessment practices rather than logistics and management.

...[modern assessment] systems allow practitioners to **focus on quality assessment practices rather than logistics and management.**



Common applications of group testing technologies span several categories, including:

- **Speed and Agility Assessment:** [Timing gates](#) such as SmartSpeed Plus support engaging, quick assessments of sprint and change-of-direction (COD) performance.
- **Force and Power Assessment:** Technologies like ForceDecks or SmartJump allow rapid strength, power and jump screening.
- **Isometric Strength Testing:** Fixed-frame and [handheld dynamometry](#) technologies, such as [ForceFrame](#) and [DynaMo](#), are used to assess joint- and muscle-specific force capacity.
- **Movement Assessment:** Markerless motion capture systems such as HumanTrak provide efficient analysis of movement strategy and range of motion (ROM).

Selecting the **Right Technology**

Not all testing systems perform well in group environments. Tools that lack validation or reliability can introduce noise, reduce comparability and weaken decision-making, particularly when data is collected at scale. For practitioners running group testing across teams, workplaces, clinics or academic settings, technology must support efficient setup, smooth participant flow and consistent data capture.



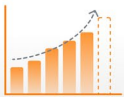
Reliability

Systems should produce consistent, repeatable measurements across participants and sessions. Reliability is a key component of practitioner confidence in data-informed decision making.



Efficiency

Automated data capture and background processing reduce testing bottlenecks. Minimizing manual recording and analysis helps maintain athlete flow and preserves data quality under time constraints.



Scalability

Technology should function effectively across varying group sizes. The ability to run multiple devices simultaneously and consolidate results into a centralized platform supports efficient review, reporting and longitudinal monitoring.

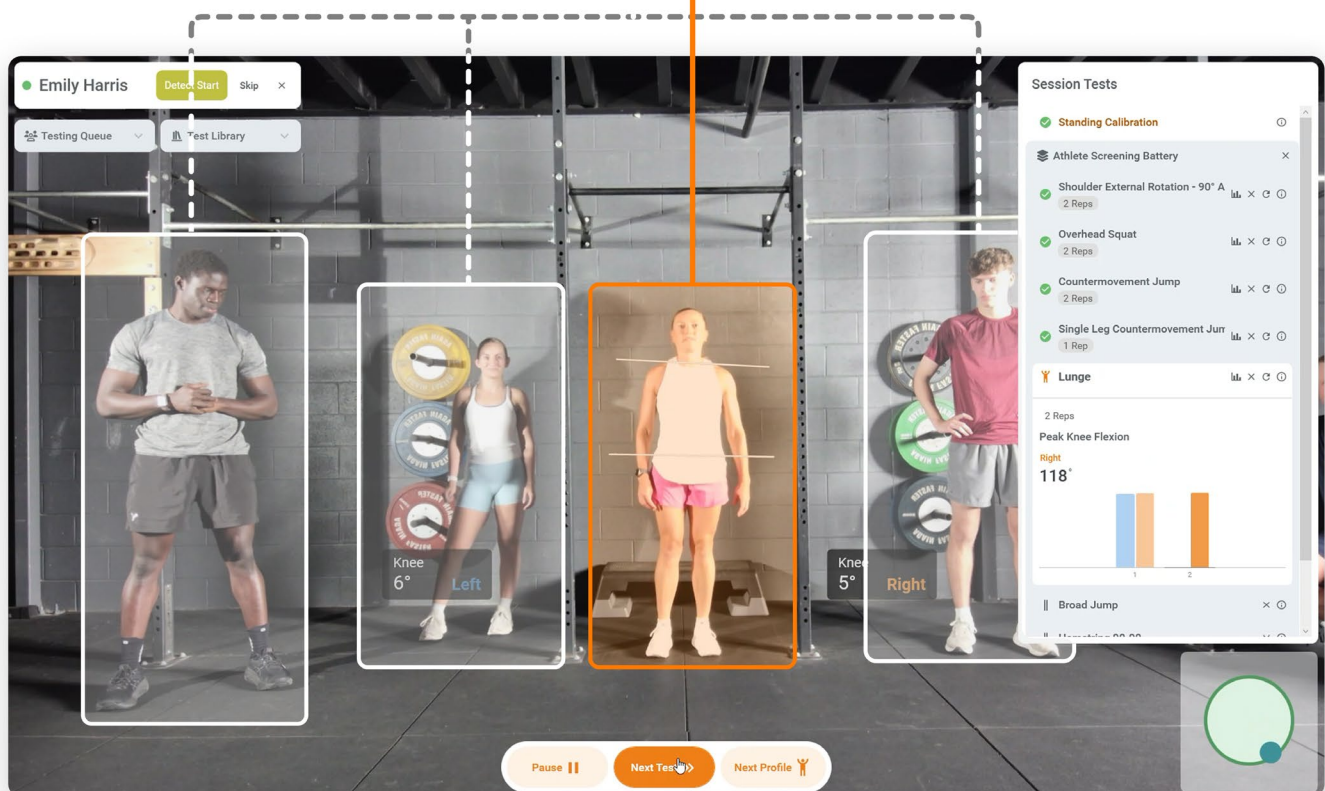
[Practitioners should prioritize] systems that **combine proven reliability with features designed for high-volume use...**

In practice, this means prioritizing systems that combine proven reliability with features designed for high-volume use, such as rapid setup, automated data handling and support for multiple participants. Choosing technologies with these capabilities helps maintain data quality while allowing group testing sessions to run efficiently and predictably.

High-Efficiency **Movement Screening**

Motion capture systems add the most value to practitioners when they minimize setup times and reduce time loss between assessments. Technologies such as HumanTrak optimize efficiency through group testing capabilities, allowing participant queues and test batteries to be linked in advance to support high-volume testing scenarios.

Multiple participants in frame
with **only one being assessed**



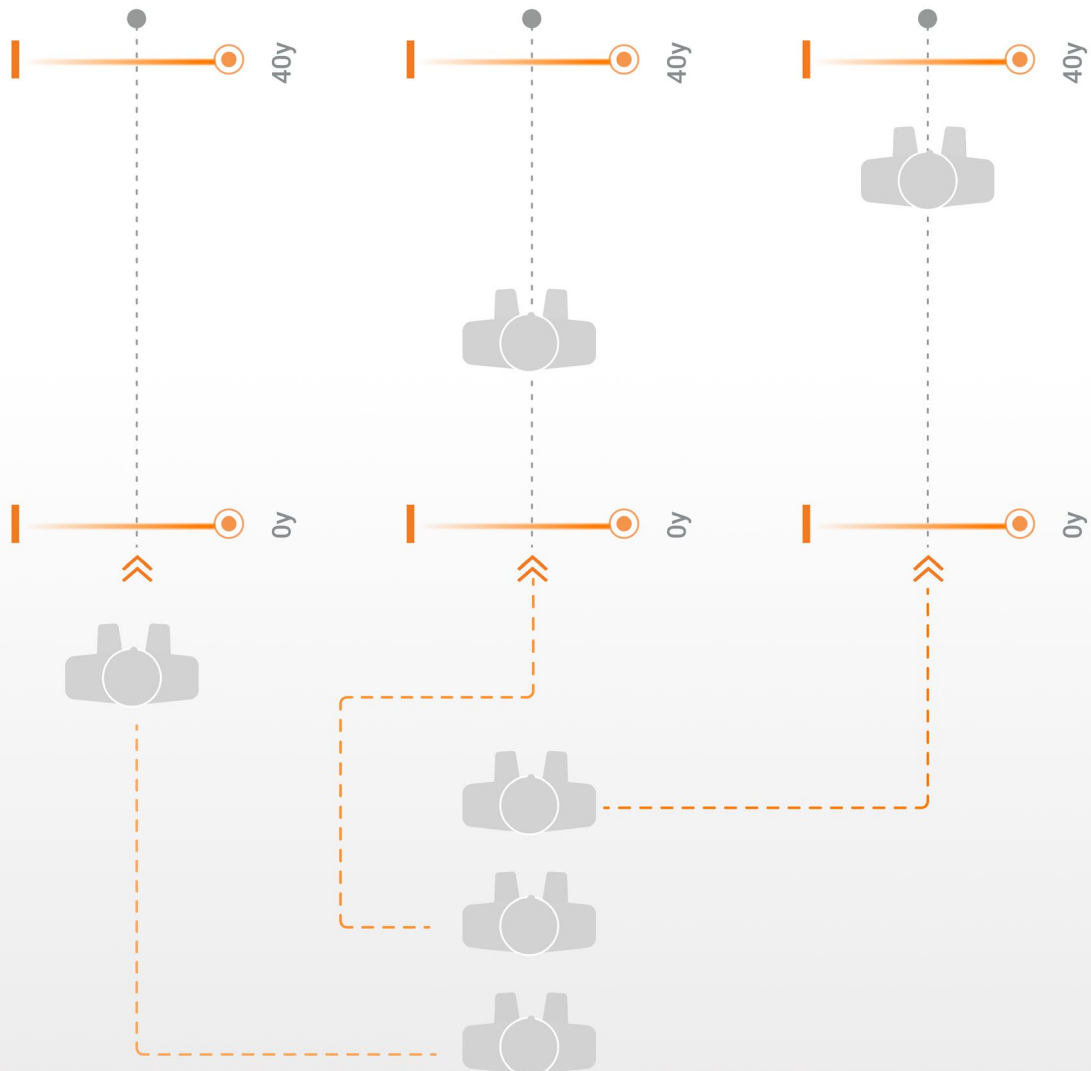
In combine-style testing sessions where jump performance is a priority, 20 participants can be assessed in under 20 minutes with features with features that enable high-efficiency group testing.

Calibration of motion capture technologies is often a bottleneck in high-output testing scenarios, leading to extended time delays during group testing. However, modern technologies, such as HumanTrak, allow a single calibration to be used across multiple dynamic tests, reducing downtime while repetition detection, visual feedback and repetition editing help maintain consistency and data integrity.

Multi-Track **Timing Assessment**

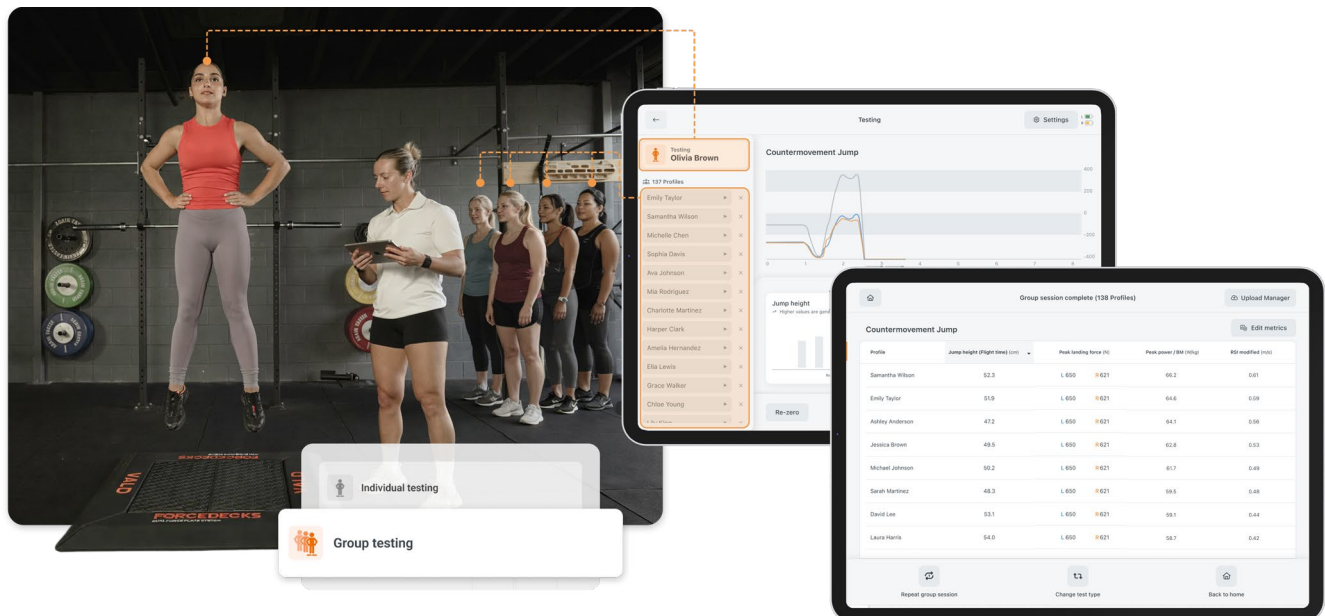
Speed assessments are meant to be efficient and simple. However, high-volume testing environments make single-track testing options slow and magnify downtime for participants. Having multi-track assessment options, as provided in SmartSpeed Plus, allows practitioners to test multiple participants simultaneously, optimizing efficiency and increasing engagement during group testing.

Having multi-track assessment options...**[optimizes] efficiency and [increases] engagement during group testing.**



High-Volume **Force Plate Assessment** (Coming Soon)

Force plate assessments are among the most commonly used tests in group settings, where practitioners benefit from workflows that allow quick selection of preset groups, easy switching between participants and clear group-level reporting within the application. An upcoming ForceDecks update will introduce features aligned with these needs, further supporting efficient and scalable group testing workflows.



Together, group-enabled workflows in systems such as HumanTrak, SmartSpeed and ForceDecks allow practitioners to scale assessments without compromising execution, making them well-suited for combines, academies, workplace screenings and other high-volume testing scenarios.

Selecting the **Right Tests**

Selecting the right tests for a group testing session begins well before equipment is set up. Effective test selection involves reverse engineering the task, identifying the physical qualities that underpin performance, health or injury risk and choosing assessments that directly reflect those qualities.

Effective test selection involves reverse engineering the task, identifying the physical qualities...and choosing assessments that directly reflect those qualities.

Rather than defaulting to familiar or generic tests and attempting to extract multiple, unrelated metrics, practitioners should be intentional in selecting tests that align with the specific outcomes they want to monitor or influence.



Across settings, this process remains consistent, even though the tasks and priorities differ:

- **Sport Monitoring:** Start with the demands of training and competition. When monitoring neuromuscular status, select tests that are simple to execute, sensitive to change and accumulate little fatigue, as monitoring assessments are frequently conducted to assess participant readiness. Though sport-dependent, common monitoring assessments are CMJs and joint-specific isometric tests (e.g., [knee extension](#) or [shoulder external and internal rotation \[ER/IR\]](#)).
- **Workplace Wellness and Injury Screening:** Begin with job-specific task demands by integrating tests that reflect the physical exposures most closely linked to injury risk or workplace demands. Manual handling roles may warrant upper-limb or trunk strength assessments (e.g., [Athletic Shoulder \[ASH\] Test](#) or [isometric mid-thigh pull \[IMTP\]](#)), while prolonged standing or repetitive work may place greater emphasis on lower-limb capacity or balance (e.g., [single leg stand](#)).
- **Combines and Talent Identification:** Work backward from the sport or positional demands. Tests are often selected to capture foundational qualities such as maximal force, sprint capacity or power (e.g., flying 10m sprint or [single leg jump](#)), which are known to underpin long-term performance potential rather than short-term readiness.

Across all scenarios, test selection is how practitioners optimize the measurement of key qualities. A focused test battery improves efficiency, reduces noise and ensures group testing data remains interpretable and actionable.

A focused test battery improves efficiency, reduces noise and **ensures group testing data remains interpretable and actionable.**

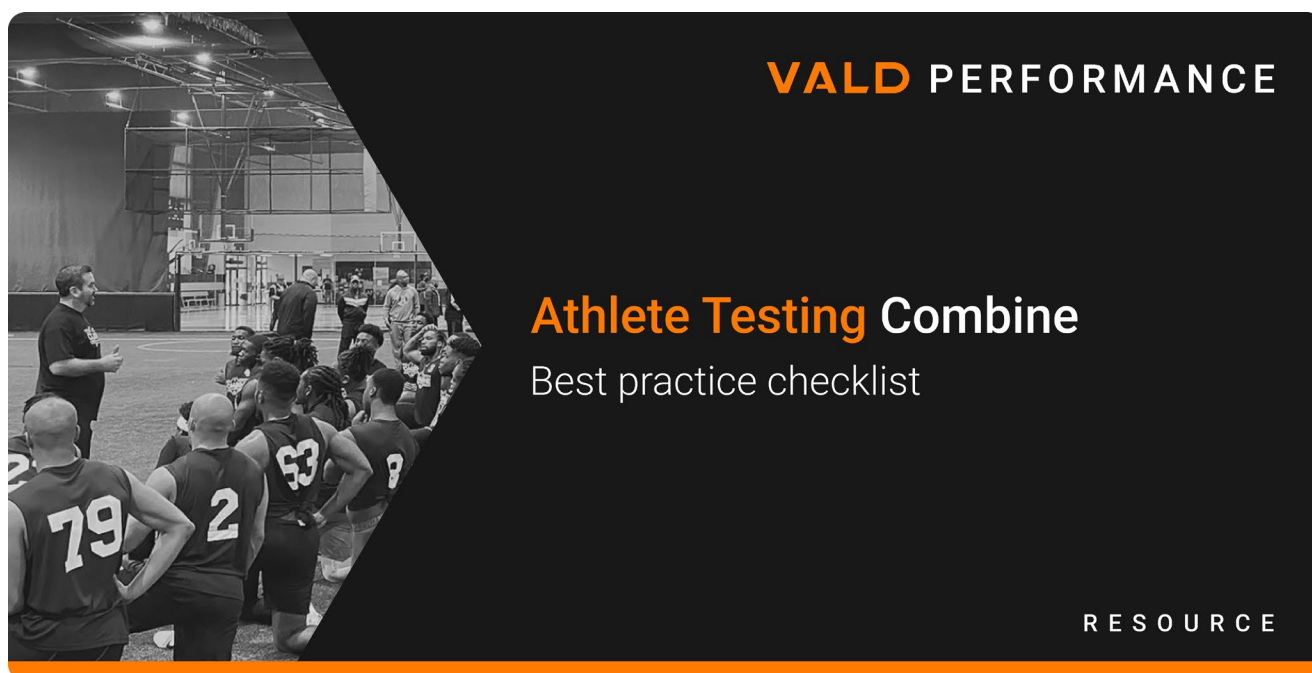
For a starting list of simple tests and metrics suitable for a generic combine assessment battery, see the [Additional Resources](#) section.

Pre-Event Preparation

The success of a group testing session is determined by the practitioner's preparation for the event. When the venue, staff and equipment are prepared in advance, practitioners can focus on test execution, participant management and clear communication rather than troubleshooting on the floor.

When [logistics]...are prepared in advance, practitioners can focus on **test execution, participant management and clear communication...**

Practitioners can reference the [Athlete Testing Combine Best Practice Checklist](#) to follow the exact steps necessary for running successful group testing events.



Planning and Logistics

Effective planning and logistics underpin successful group testing, ensuring that venues, testing order and equipment are aligned to support smooth participant flow, minimize fatigue-related variability and maintain data quality at scale.

Venue Selection and Layout

Choose a space – if you have the choice to – that supports smooth participant flow with minimal interference between stations. To support this, consider the following layout guidelines:

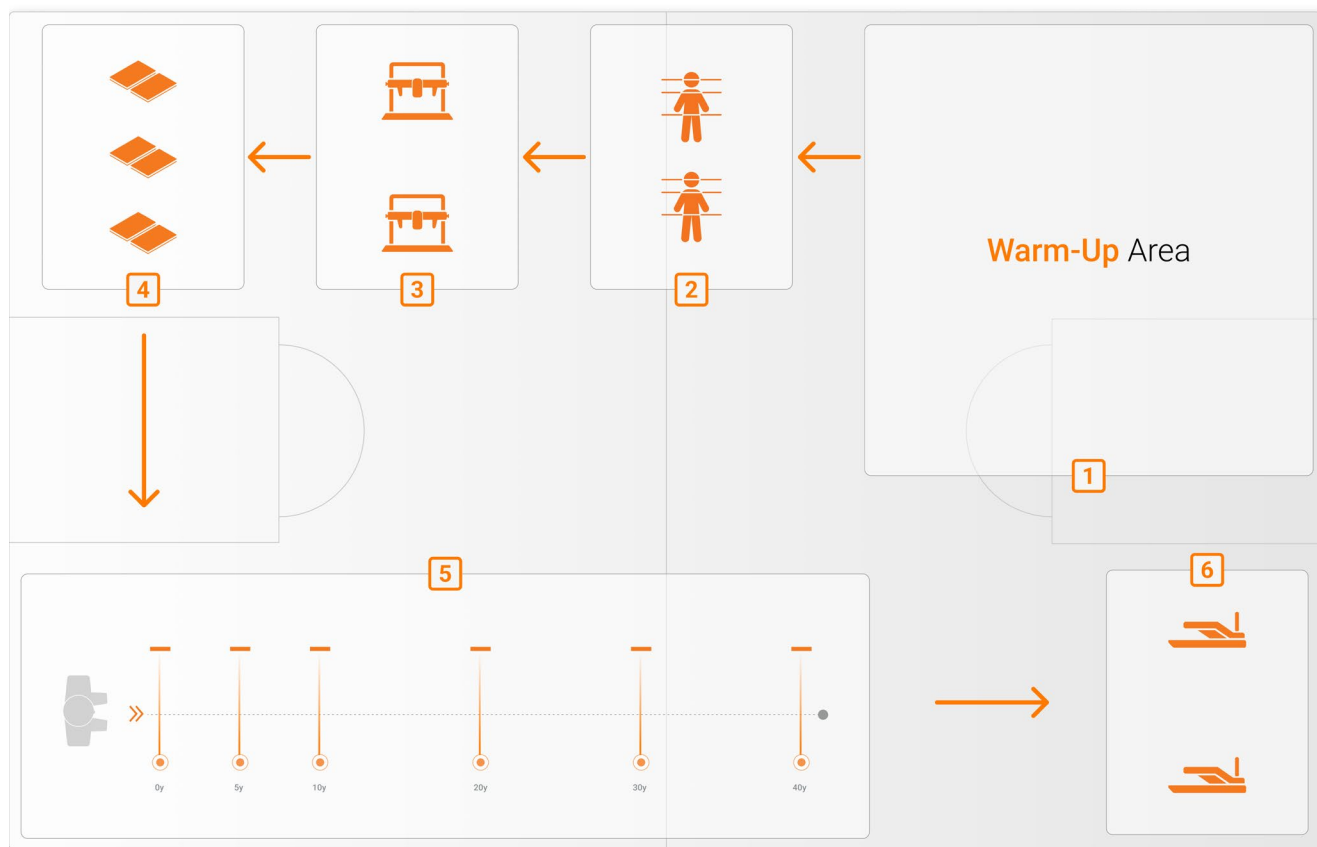
- **Sprint assessments** should be positioned in open, obstruction-free lanes.
- **Jump and isometric testing** work best in quieter zones where participants can focus and where data capture is not disrupted by external movement or vibration.
- **Circular or linear flow** should be established so participants can move logically from one test to the next with minimal downtime.



Testing Order

Plan the testing order with the physiological demands of each test in mind to minimize fatigue-related variability across assessments. For example, jumps or isometric tests are typically completed before sprints or other high-intensity drills. The following sequence demonstrates a simple flow practitioners can use as a starting point, recognizing that test prioritization may vary depending on the testing population and performance demands.

Plan the testing order with the physiological demands of each test in mind to **minimize fatigue-related variability across assessments.**



1. **Warm-Up**
2. **Movement and ROM Assessments:** Assess joint mobility and ROM before maximal effort isometrics, jumps and sprints as they can significantly alter joint motion.
3. **Isometric Assessments:** High peak force with low total strain or fatigue accumulated. It is important to consider that potentiation effects from one test may influence the performance of other assessments (e.g., high IMTP efforts may potentiate adduction and abduction [Add/Abd] performance if one is performed immediately after the other).
4. **Jump Assessments:** High-effort dynamic movements that are simple and quick to assess. Accumulate reasonably low fatigue.
5. **Sprint and COD Assessments:** Maximal effort, high-speed movements that likely induce higher levels of fatigue, specifically in the central nervous system and lower extremities.
6. **High-Fatigue Assessments:** Eccentric testing (e.g., [Nordics](#)), repeated sprint ability (RSA) or other maximally fatiguing assessments should come last in a group testing event to avoid affecting the performance of other tests.

Similar sample testing schedules, flowcharts and resources can be found at the end of this guide in the [Additional Resources](#) section.

Equipment Checklist

Before the event, confirm that all required systems are functional and ready to use. Taking time to turn systems on, sync connections (e.g., Bluetooth and Wi-Fi) and walk through system procedures helps prevent delays and ensures a smooth testing experience for all participants. Practitioners should complete the following tasks:

- **Fully charge** all devices and test battery levels.
- **Pack accessories** such as tripods, cables and additional batteries.
- **Close and secure travel bags** and protective cases before transporting.

For large-scale events such as draft combines or academy screenings, resources like [VALD's Athlete Testing Combine Best Practice Checklist](#) help ensure that practitioners do not miss critical setup steps, supporting consistent testing across staff, stations and locations.

Athlete Testing Combine Best Practice Checklist

STAGE 4 - POST-EVENT
☐

STAGE 3 - EVENT DAY
☐

STAGE 2 - PREPARATION (1 WEEK - 1 DAY BEFORE EVENT)
☐

STAGE 1 - PLANNING (2-8 WEEKS BEFORE EVENT)
☐ Define Testing Objectives
☐ Select Testing Protocols
☐ Procure Testing Systems
☐ Schedule Testing Dates and Venue

Registration and Communication

Clear pre-event communication sets expectations and reduces delays on the day of testing. A concise message sent in advance should include the following information:

- **Testing details**, including date, time and location
- **Required attire**, such as clothing or footwear
- **Expected effort level**, such as maximal testing or familiarization
- **Pre-test considerations**, including activity restrictions or nutritional timing
- **Registration instructions** for data-access platforms like [MoveHealth](#)

Providing this information ahead of time supports smoother session flow and more consistent testing conditions. Templates for participant messages, checklists and other group testing-related resources can be found in the [Additional Resources](#) section of the guide.

Profile Management

Efficient registration and profile management before testing help reduce on-site delays and ensure participants move through assessments smoothly. Uploading participant rosters into testing ecosystems such as VALD Hub before the event helps eliminate delays and allows practitioners to assign groups or tags if needed.

Efficient registration and profile management before testing help reduce on-site delays and **ensure participants move through assessments smoothly.**



Expert Tip: Bulk Imports

For large groups, bulk import features simplify setup – ensuring each participant is pre-registered and ready to be assigned to tests. If additional metadata (e.g., sport, position, injury history) is required, **practitioners can use the bulk upload templates provided on VALD Knowledge Base.**

Bulk Upload

RB Ryan Baugus
Headquarters Physical Therapy

1

Upload CSV

2

Review Profiles

3

Import Profiles

Required Fields:

Family Name

Given Name

Date of Birth

Sex

Additional Fields:

Email

Weight

Height

Sport

Position

Categories

Tags

Photo and Video Permission

The bulk upload feature in VALD Hub distinguishes between required and additional fields to give practitioners the option of adding specific attributes about each participant.

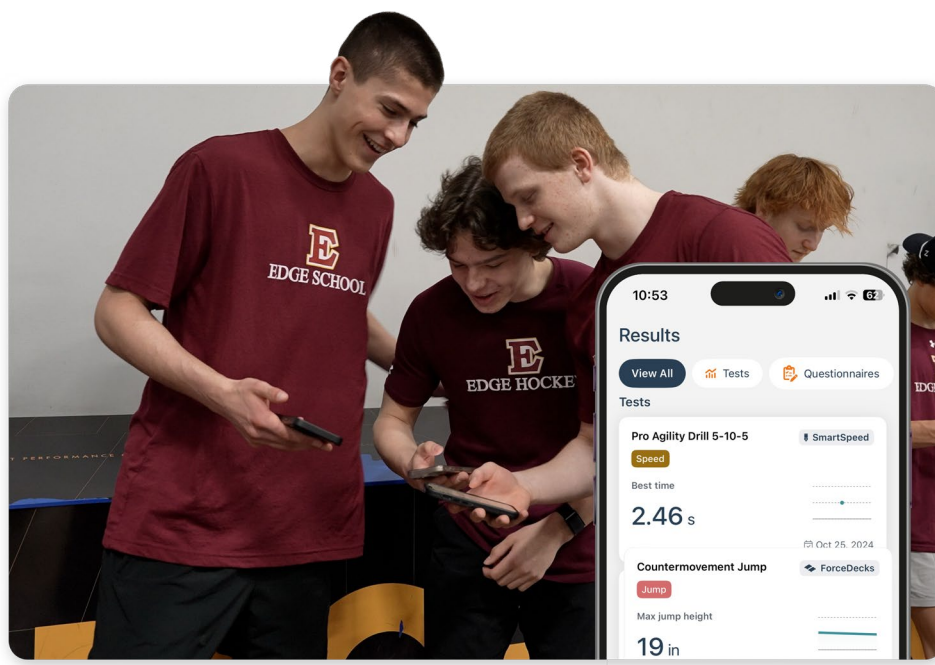
Once participant profiles have been uploaded into a testing portal (e.g., [VALD Hub](#)), invitations to public-facing applications such as MoveHealth can be sent to make the data-sharing process simple and effective.

Data Distribution

Once results are uploaded into a data management ecosystem like VALD Hub, practitioners can explore how data can be automatically distributed to participants in an efficient and effective manner. [Digital health](#) platforms such as [MoveHealth](#) require minimal setup, offer free access and are often embedded within these systems. Adding a participant's email address allows practitioners to send app invitations, with results automatically linked and available as soon as data is uploaded.

...MoveHealth, [requires] **minimal setup, [offers] free access and [is] embedded within [VALD Hub].**

For example, leading up to a combine or group testing day, participants can create profiles via [invite links](#) to streamline the data distribution process.



Normative data comparisons are another high-value addition to data distribution. Databases such as VALD Norms automatically sync with MoveHealth to provide participants with age- and sex-matched normative data comparisons, helping them understand how they stack up against their peers.

...[VALD Norms] provide participants with age- and sex-matched normative data comparisons, **helping them understand how they stack up against their peers.**

For practitioners, these platforms also eliminate the need for manual reporting while delivering scalable, consistent and easy-to-understand feedback across group testing environments.

Consent and Privacy

Clear data governance is essential when conducting group testing in schools, clinics and public-facing environments. Practitioners must ensure consent, security and data access are managed appropriately while maintaining long-term data value.

Practitioners must **ensure consent, security and data access** are managed appropriately while maintaining long-term data value.

Compliance and Security

Group testing events commonly use waivers, consent forms and data-sharing agreements to help ensure safe and legal release of information. However, these tactics lack the sophistication needed to keep health and performance information secure.

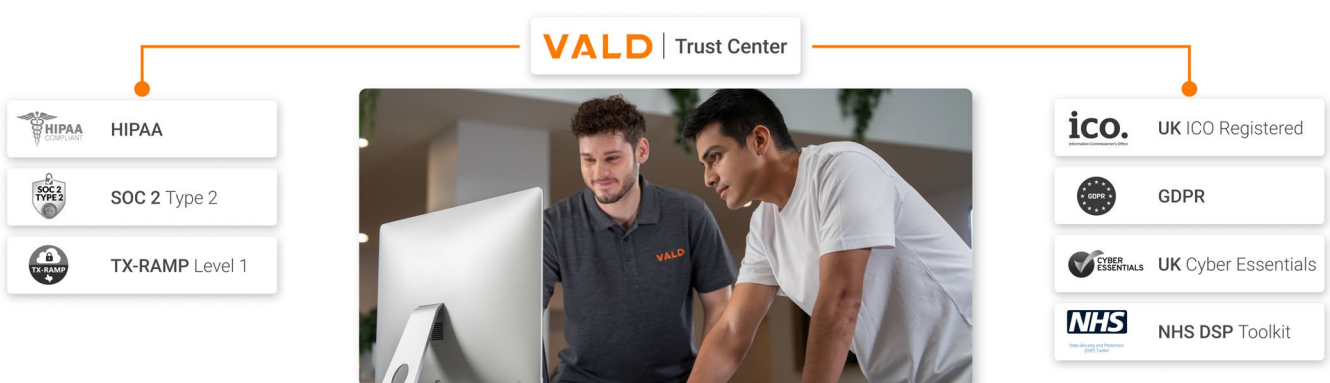
Data security and compliance standards, such as the [Health Insurance Portability and Accountability Act \(HIPAA\)](#), [SOC 2 Type 2](#) and the [General Data Protection Regulation \(GDPR\)](#), are critical for practitioners running group testing events at any scale, as data collected at these events can be considered personal health information (PHI). Practitioners should ensure the testing ecosystem they use (e.g., VALD systems, MoveHealth and VALD Hub) is secure and meets data protection standards.

Practitioners also remain responsible for maintaining security compliance when sharing data outside of testing ecosystems, including through email, SMS, social media and external analytics platforms (e.g., Excel or Power BI).

Access Control and Long-Term Tracking

User permissions and system access should be governed by managers or organization leaders, tailoring access levels to specific providers. For example, VALD Hub allows practitioners to control who can view, edit or export data, providing a secure platform based on practitioner login.

Storage of participant data should support both individualized analysis and trend comparisons. Some digital ecosystems, such as VALD Hub, provide [Timeline](#) views and [Group Dashboard](#) features beyond typical participant profiles to support trend- and group-related reporting.

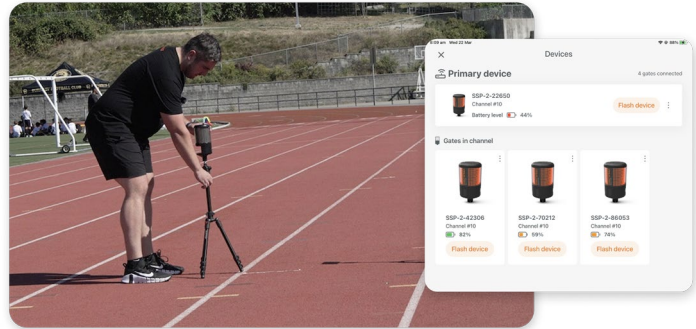


Together, these features support safe, compliant and scalable data management while preserving the long-term value of group testing outcomes.

Technology Setup

Proper technology setup is essential for efficient group testing and reliable data collection. Practitioners are encouraged to arrive with sufficient time before the session to complete a simple checklist, ensuring system-specific considerations are addressed. This preparation reduces disruptions and supports consistent data collection once participants arrive.

...preparation reduces disruptions and supports consistent data collection once participants arrive.



The following checklist outlines key setup steps for common testing systems:

- **Timing Systems:** Align timing gates and confirm sensor height and spacing are consistent across lanes.
- **Force Plates:** Level and zero plates to ensure they are ready for testing.*
- **Strength Testing Systems:** Confirm dynamometry setups (e.g., ForceFrame crossbar positioning) match the planned assessments, such as hamstring, adductor or shoulder testing.
- **Software Checks:** Ensure software is up to date and results screens display the correct metrics to optimize data transfer.
- **System Checks:** Complete one or two test trials to confirm that all systems are connecting, syncing to VALD Hub and displaying data as expected.

* Practitioners using force plates on a non-rigid surface can bring a portable, rigid testing surface (e.g., a hard plywood sheet, metal plates or thick gym flooring) to ensure clean, comparable data is collected.



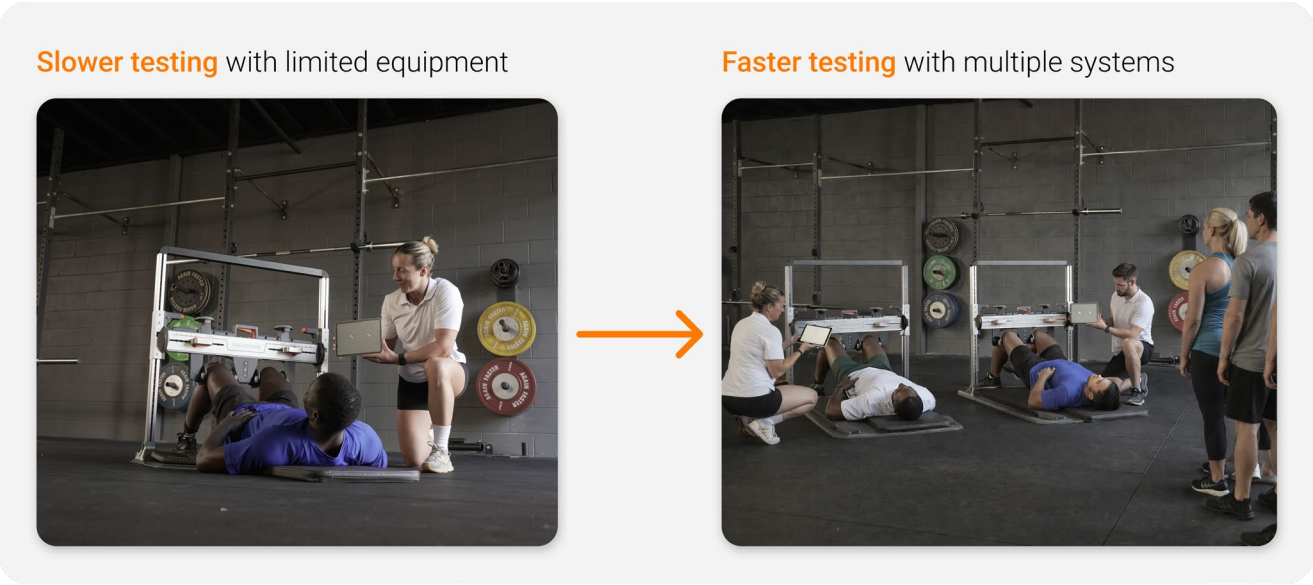
Rigid surface used to ensure testing consistency and **accurate data collection**

Common Pitfalls and Solutions

Many issues in group testing stem from preventable setup and workflow challenges rather than the testing itself.

Bottlenecks commonly occur around tests that require longer setup, participant-specific adjustments and potential familiarization repetitions. For example, isometric hip adduction and abduction (Add/ Abd) assessments often take extra time for practitioners to match crossbar height to the desired hip flexion angle (e.g., 45° or 60°) based on participant anthropometrics.

Bottlenecks [include]... **longer setup, participant-specific adjustments and potential familiarization repetitions.**



The following table provides simple solutions and rationale for common group testing bottlenecks:

Bottleneck	Solution	Rationale
Long Assessment Time per Test or System	Add systems or tracks to each station	Allows multiple participants to be tested at once
Long Changeover Time Between Participants	Group participants based on anthropometrics	Minimizes system adjustments between participants, speeding up testing
Inconsistent Work and Rest Periods	Pair complementary tests together	Expedites testing and assesses similar qualities simultaneously

Technology-related disruptions are sometimes unavoidable and pose a risk in high-volume testing environments. The following list provides solutions for potential technology issues, such as connectivity and power.

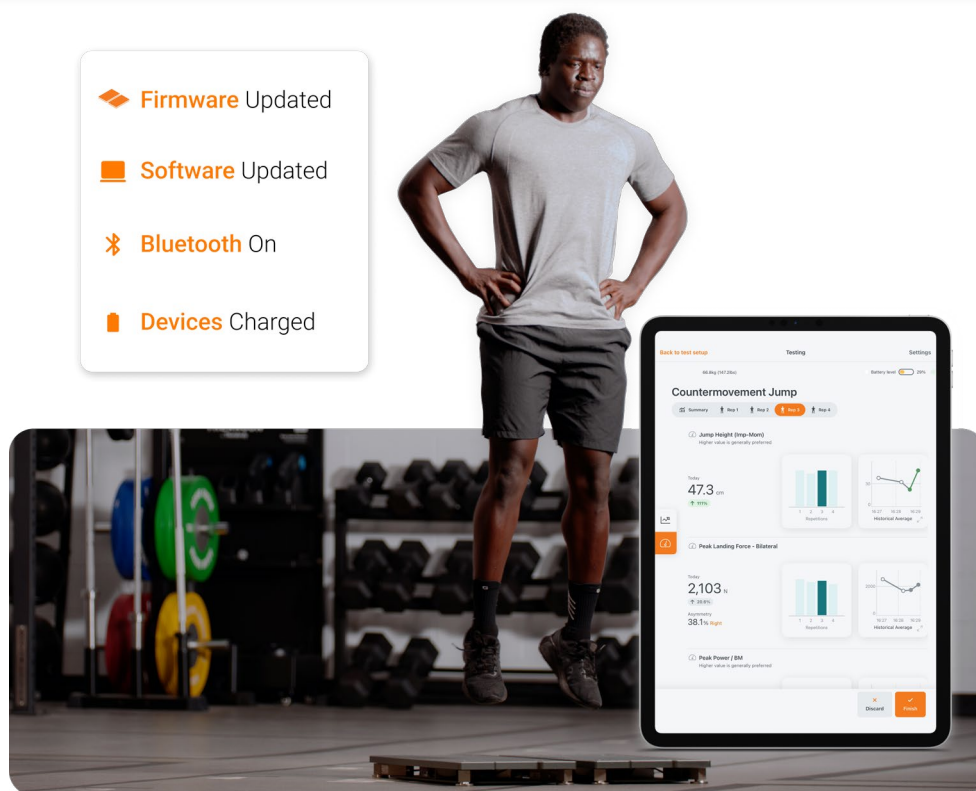
Connectivity

- Ensure all applications are up to date, measurement technology firmware is current and phone or tablet software is up to date prior to testing.
- Use systems that can collect data offline and automatically upload data when reconnected (e.g., VALD technologies).*
- Provide Wi-Fi hotspots through designated devices or smartphones during the testing session.
- Perform one sample test on each system during setup to ensure everything is running smoothly.

Power

- Charge all systems the night before an event.
- Bring spare batteries (if available) and charging cables in case of power supply issues.

* You cannot log in to the iOS apps without an internet connection. You must log in while connected to the internet before attempting to conduct tests offline.



Basic technology checklist to ensure a smooth technology-enabled testing experience.

Finally, staff communication and adaptability play a central role in successful group testing. Sessions run most effectively when teams prioritize readiness and execution quality over rigid adherence to a schedule. This means letting participants retest if their execution was incorrect or their results seemed out of the ordinary, even if it increases testing time at a given station by a few seconds.

...staff communication and adaptability play a central role in successful group testing.

Running the Event

Effective group testing balances smooth athlete flow with consistent execution and data quality, while maintaining a positive testing environment.

Practical strategies for event management, staff coordination and real-time data oversight help ensure testing days operate seamlessly from start to finish.



Event Flow and Station Management

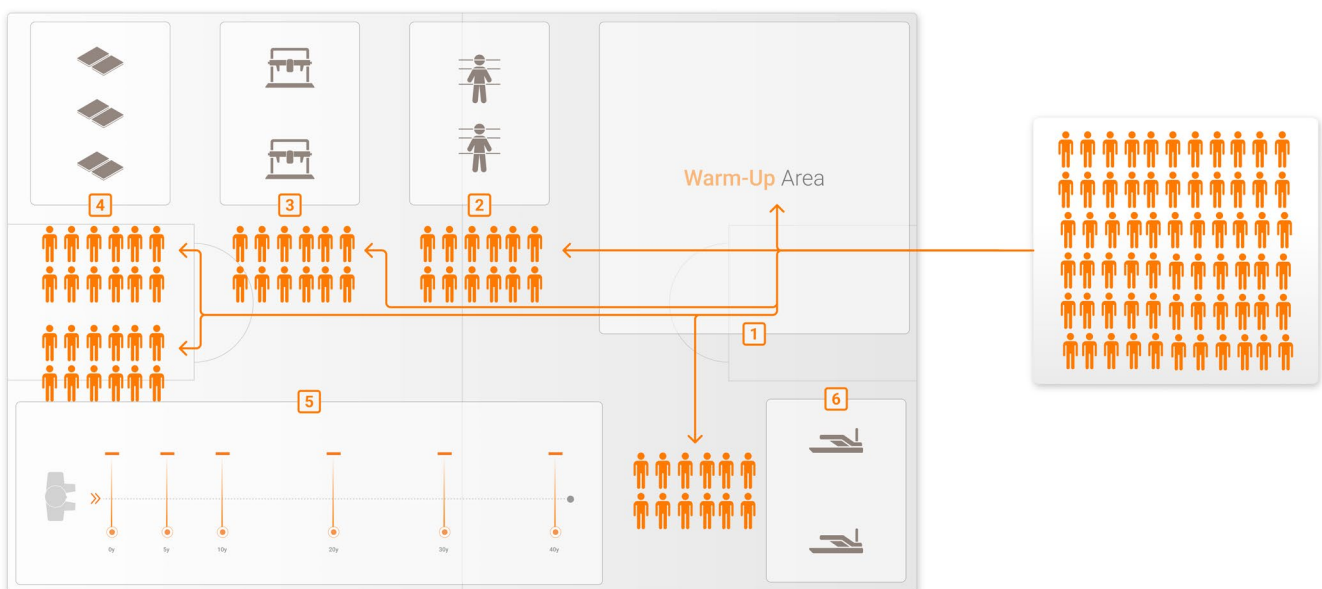
Effective group testing depends on clear station flow, coordinated timing and structured staff communication to keep participants moving while maintaining consistency and data quality.

Station Flow

A clear station flow is critical for managing large groups efficiently. Two common approaches are used, each with distinct trade-offs.

Equal Distribution

All participants are spread out equally, rotating until they have completed all stations

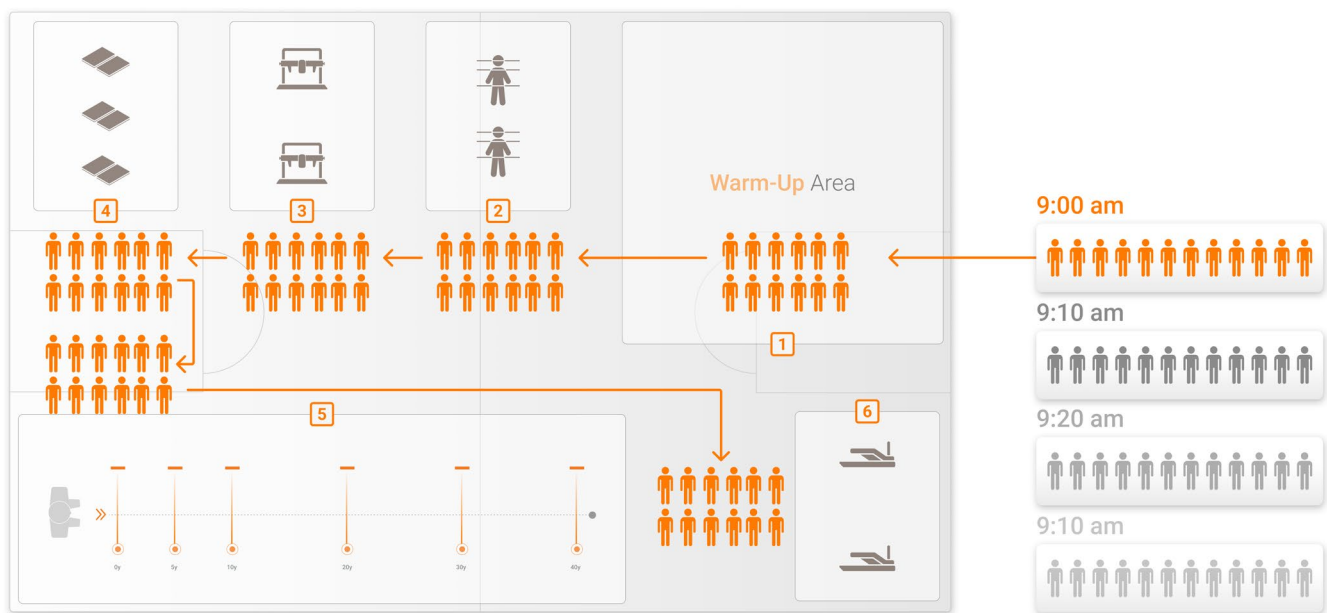


With an equal distribution approach, participants begin at the same time and are spread evenly across testing stations, rotating at set intervals once each station is complete. This method supports fast, time-bound testing sessions with a clear start and finish, but provides less control over fatigue and test order, which can limit comparability between groups.

[Equal distribution] supports fast, time-bound testing... but provides less control over fatigue and test order...

Staggered Distribution

Participant groups arrive at staggered times to maintain the testing sequence



With a staggered distribution approach, participants are pre-assigned to groups with scheduled arrival times and progress through the same sequence of tests. This improves consistency of test order and fatigue exposure across groups, supporting cleaner comparisons, but typically requires more coordination and logistical planning to execute smoothly.

[Staggered distribution] improves consistency of test order and fatigue exposure... but typically requires more coordination and logistical planning to execute smoothly.

When selecting a staggered distribution approach, participants should be grouped by skill level, anthropometrics or testing experience to help facilitate a more seamless testing process.

Timing and **Scheduling**

Clear timing and scheduling are essential for maintaining session flow, minimizing delays and preserving data quality during group testing. Practitioners can ensure a smooth testing event with strategies like the following:

- Using a **central timer or clock** visible to all staff
- Accounting for **short transition periods** (1-2 minutes) between stations
- Adding **buffer time** to allow for occasional retesting, incorrect timing assumptions and uncontrollable changes (e.g., bathroom breaks)
- Scheduling **brief breaks for practitioners** every 45-60 minutes to maintain focus and testing rigor

Clear timing and scheduling are essential for **maintaining session flow, minimizing delays and preserving data quality** during group testing.

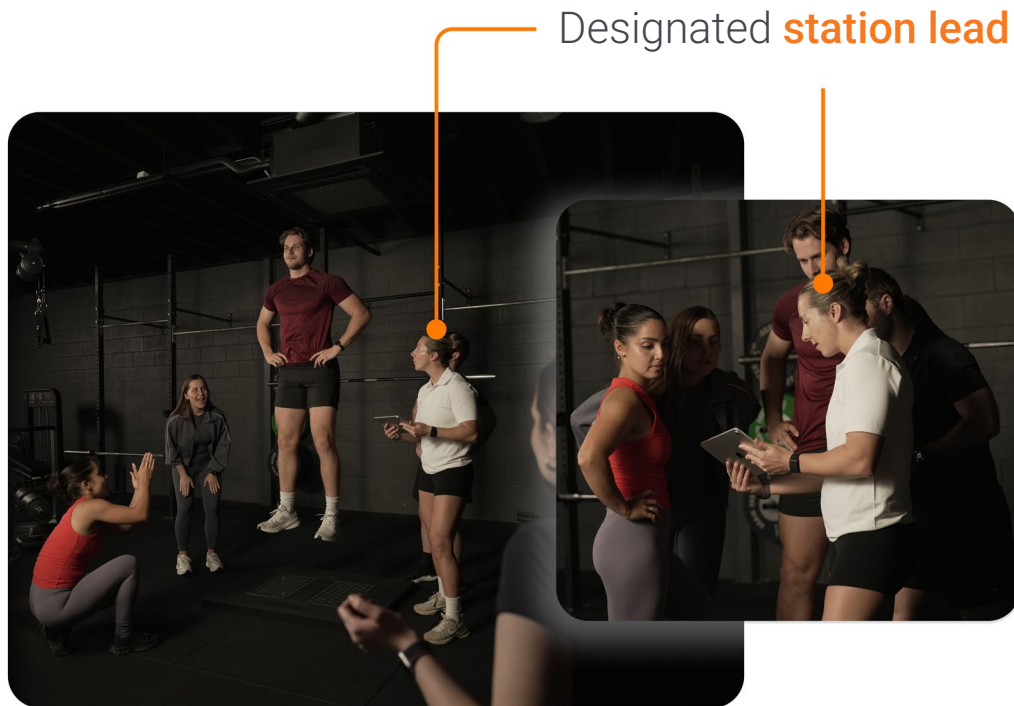


Designated station lead coordinating group dynamometry testing, adherent to timing and scheduling.

Establishing a clear schedule and adherence strategy keeps participant testing on time, allowing practitioners and testing leads to focus on data quality rather than logistics.

Staff Communication and Coordination

Each testing station may be assigned a designated lead to manage instruction and data capture, while a central coordinator often oversees timing, station rotations and troubleshooting.



Designated station lead conducting testing and sharing performance results with the group.

Communication can be supported through radios or headsets in larger venues, while verbal cues or hand signals are likely sufficient in smaller setups. Consistent communication supports smooth transitions and helps minimize bottlenecks throughout the session.

Consistent communication supports smooth transitions and helps minimize bottlenecks throughout the session.

For more information about efficient practitioner communication and engagement, see the [Additional Resources](#) section.

Preparing for Testing

Preparing participants appropriately before testing helps reduce testing variability, improve execution quality and ensure group testing sessions run efficiently from the first assessment.

Warm-Up Protocols

Standardizing warm-ups across the group helps reduce variability between participants, ensuring that differences in participant testing data are more likely due to differences in speed, strength or endurance, rather than noise. A typical group warm-up might include the following:

- **General dynamic warm-up** (3-7 minutes)
- **Specific mobility or activation drills** (2-3 minutes)
- **Submaximal familiarization repetitions** for assessments (e.g., CMJs, isometrics or short sprints)



Standardizing warm-ups...helps reduce variability between participants, ensuring that differences in participant testing data are more likely due to [true performance differences]...

Similarly, practitioners may incorporate submaximal assessment trials throughout the rotations as both a warm-up drill and a familiarization method for participants who may be new to assessments such as hands-on-hips jumping or isometric testing.

Participant Briefing

Before testing begins, bring all participants together for a short briefing outlining the following:

- **The purpose** of the testing session
- **The order** of tests and rotation flow
- The key **performance cues** (e.g., “jump as high and as fast as possible”)



A short demonstration of each test helps ensure participants understand the correct technique. This also allows staff to confirm that all systems are functioning before rotations begin.

After a set time (e.g., 10 minutes), groups rotate to the next station. Structured rotations maintain tempo, reduce idle time and prevent congestion at specific testing areas.

Structured rotations maintain tempo, reduce idle time and prevent congestion at specific testing areas.

Consistent Testing **Protocols**

Objectivity begins with standardization. Each participant should receive the same test instructions, verbal cues and rest intervals. Even with automated systems such as ForceDecks and SmartSpeed Plus, participant consistency remains critical.

The following are standard instructions and cues commonly used:

- **CMJ:** “Jump as high and as fast as possible.”
 - This cue maximizes jump height while minimizing contraction time.
- **IMTP:** “Pull as hard and as fast as you can.”
 - This instruction ensures that both peak- and rate-derived qualities can be accurately tracked.
- **Sprint Test:** “Run through the gates – do not slow down before the finish line.”
 - Unintentional deceleration before clearing the final gate will increase total sprint time.



Maximizing Efficiency

Practitioners experienced in group assessment recognize that testing flow and efficiency affect more than data collection. Well-run sessions support event success, athlete experience and long-term business opportunities, making high-efficiency testing strategies a worthwhile investment.

Well-run sessions support event success, athlete experience and long-term business opportunities, making high-efficiency testing strategies a worthwhile investment.

High-Volume Testing Strategies

Managing participant testing is critical in group environments where time is limited and large numbers must be assessed without sacrificing data quality. The following strategies help increase the number of participants tested within a given session:

- **Duplicate High-Demand Stations:** Assessments like [hip Add/Abd 60°](#) are highly valuable in sport but take time to set up due to varying leg lengths and body sizes. Therefore, adding multiple ForceFrame stations increases testing speed by allowing two participants to be tested simultaneously.
- **Pair Complementary Assessments:** Using SmartSpeed Plus with tracks to capture different assessments, or running more than one test on a single system (e.g., [CMJ](#) and [IMTP](#) on ForceDecks), can help expedite the testing process and improve session flow.



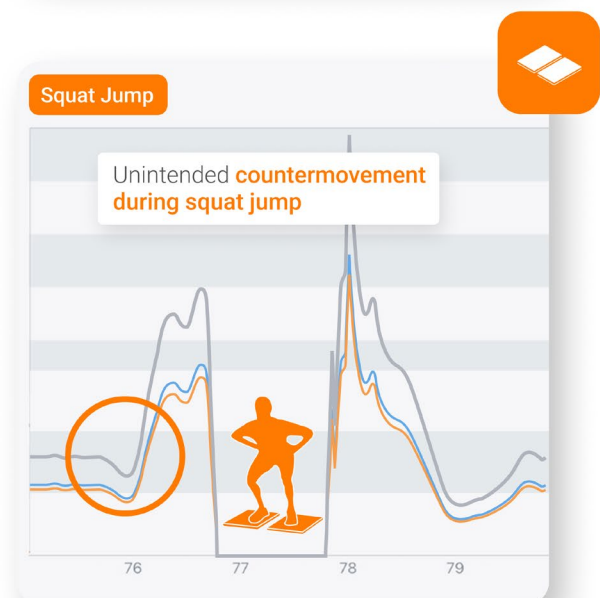
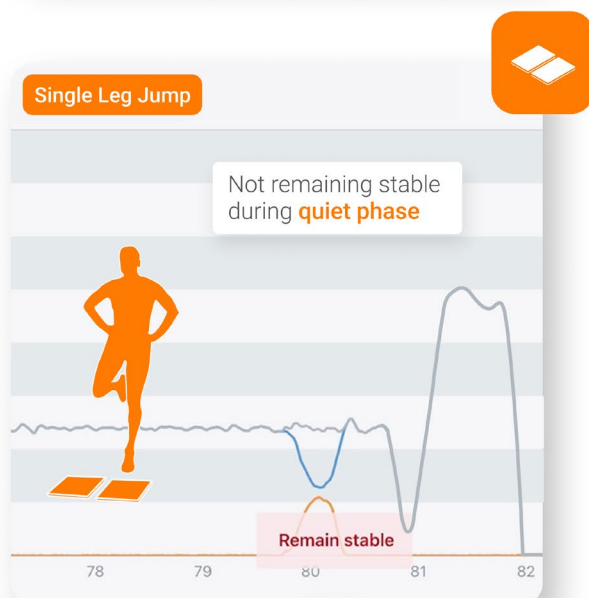
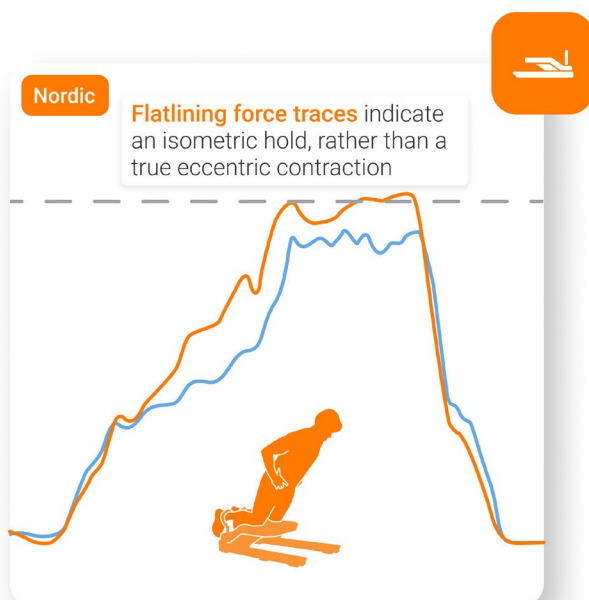
Quality Checks

Practitioners overseeing testing stations can monitor outputs and trial registrations to ensure data is recorded accurately and efficiently.

For example, when using [Auto-detect on ForceDecks](#), small unintended countermovements during a squat jump can be registered as a CMJ. In this scenario, practitioners should cue the participant to minimize any countermovement when initiating the jump, and – if needed – remove faulty tests on the upload screen in the ForceDecks app.

...small unintended countermovements during a **squat jump can be registered as a CMJ.**

The following are some common examples of potential faults that require retesting to ensure data quality:



Similarly, practitioners can use in-app results to gauge participant intent and effort during assessments. This is often done by setting movement thresholds, such as criteria for eccentric peak velocity (EPV) in a CMJ, to ensure participant effort and engagement are high.



For more insights on EPV, its applications and potential thresholds for data quality, read [The Power of EPV Part 1](#) and [Part 2](#).

Keeping Participants Engaged

Group testing can quickly feel repetitive for the participant if testing is not managed with energy and positive feedback. The following strategies can help practitioners keep participants engaged:

- Displaying leaderboards or team averages during the testing sessions (e.g., connecting a display to Leaderboards on [VALD Hub](#))
- Using in-app normative data comparisons (e.g., [VALD Norms](#))
- Providing quick verbal feedback ("That's your best jump yet!")
- Maintaining a positive and competitive atmosphere
- Encouraging participants to use secure data-sharing apps like MoveHealth to track their performance in real-time
- Providing QR codes or instructions to download digital health platforms (e.g., MoveHealth)

The instantaneous nature of real-time data visualization, as provided throughout the VALD suite of applications, helps drive motivation and effort, creating a performance-focused testing environment.

...instantaneous...data visualization...**helps drive motivation and effort,** creating a performance-focused testing environment.

More information, including testing flows and practitioner logistics planning, can be found in the [Additional Resources](#) section at the end of the guide.

Data Management and Reporting

Once testing is complete, practitioners' focus shifts from data collection to interpretation. Group testing generates large volumes of information, and its value depends on how effectively results are organized, verified and communicated.

Group [testing's]...value depends on **how effectively results are organized, verified and communicated.**

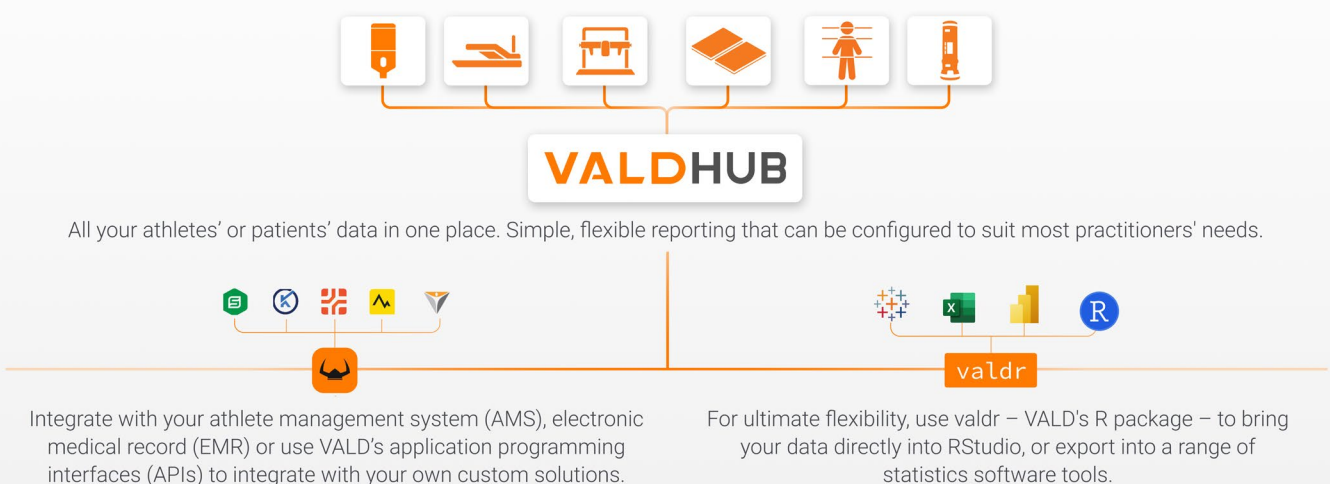
This section outlines practical approaches to consolidating data, maintaining accuracy and translating results into actionable insights while preserving participant privacy and saving time.

Data Consolidation

Modern measurement solutions reduce the need for manual file transfers and lower the risk of error during high-volume testing; however, this benefit depends on how well data from multiple systems is consolidated. In many real-world environments, objective testing platforms are used alongside other technologies, making it important to define a clear workflow for how results will be centralized, reviewed and interpreted.

Platforms such as VALD Hub streamline this process when VALD systems are used, while structured export workflows or third-party dashboards may be used when integrating data from multiple providers.

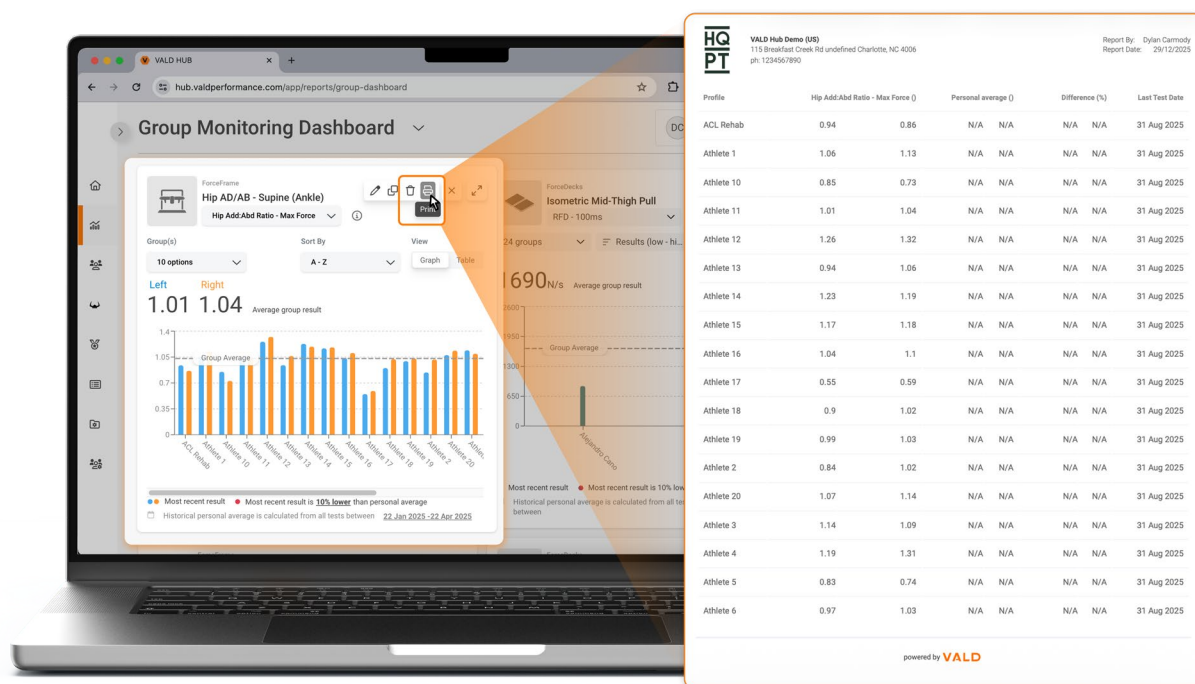
...VALD Hub [streamlines] this process...[while] third-party dashboards may be used when integrating [non-VALD] data...



For practitioners operating within broader data ecosystems, [APIs](#) provide a scalable solution for combining data across multiple measurement solutions. API-based integrations allow objective testing data to flow directly into AMS, EMR or custom dashboards, where it can be viewed alongside training load, wellness and task exposure data.

Automated Reporting

Once data has been consolidated, practitioners can leverage a range of reporting options to effectively distribute participant data. Centralized platforms enable practitioners to compare participants against group averages and available reference data, helping identify patterns, standout performances and potential areas of concern – particularly in high-volume testing environments.



Printing group reports directly from VALD Hub.

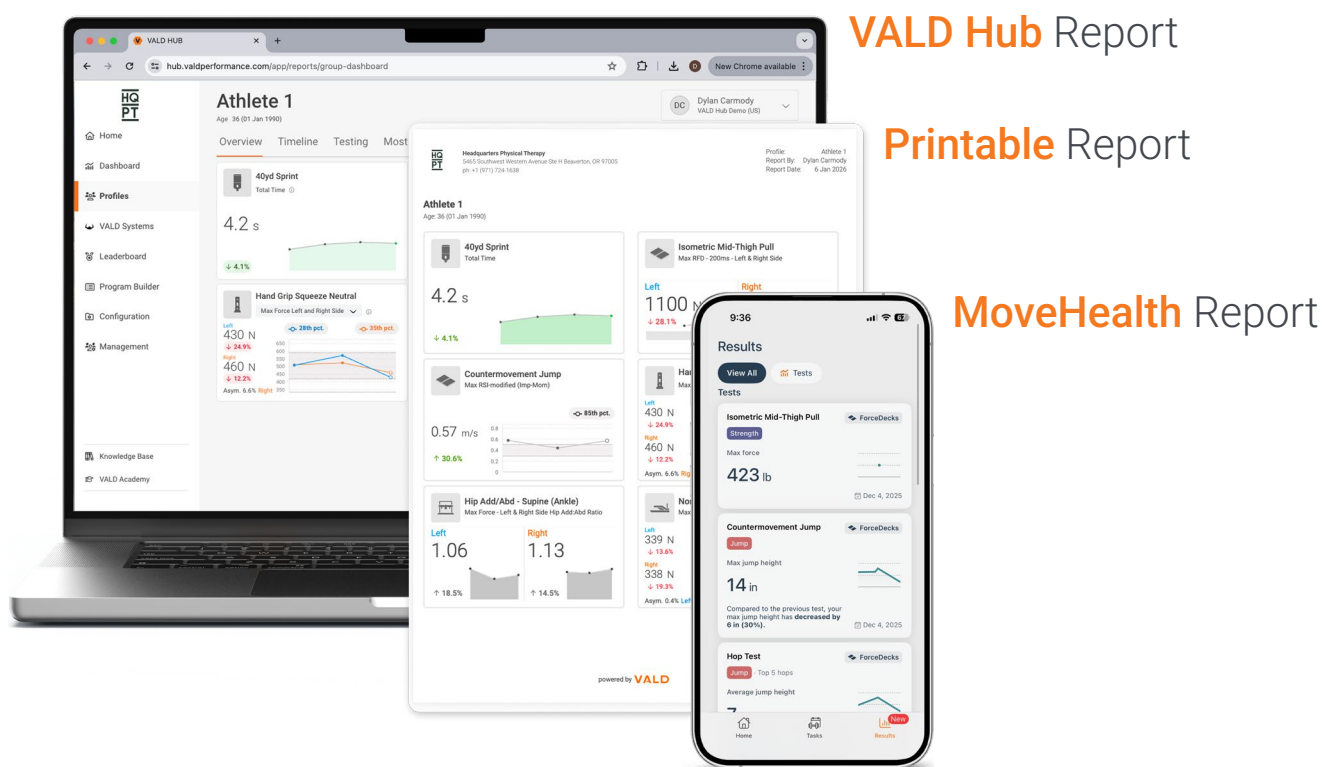
Results may be reviewed directly within platforms such as VALD Hub, shared with stakeholders or exported for further analysis, depending on the practitioner's needs.

Practitioners differ in the ways they interact with data, ranging from quick summary views to more in-depth analytical workflows. Effective reporting ecosystems (e.g., VALD Hub) support this spectrum by allowing results to be reviewed, shared and analyzed, ensuring that group testing data remains practical and interpretable.

Effective reporting ecosystems support [practitioners] by **allowing results to be reviewed, shared or analyzed [across multiple interfaces]...**

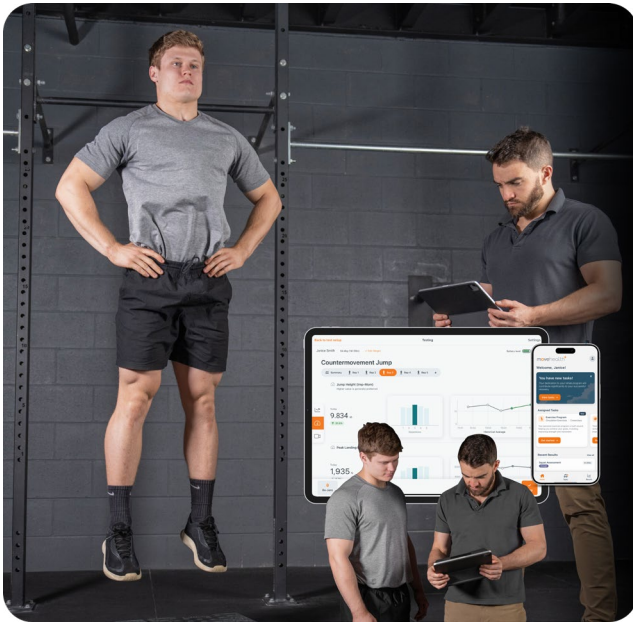
Individual Reports

Individual reporting has traditionally been time-consuming, often requiring manual exports, custom summaries and follow-up communication after testing sessions. Modern reporting workflows reduce this burden by allowing results to be delivered automatically or generated on demand, depending on the audience and context.



App-enabled platforms, such as **MoveHealth**, provide a free, client-facing option where participants can access their own results in a clear, mobile-friendly format. Results are intentionally streamlined, allowing participants to review key outcomes, compare against Norms and track changes over time without being overwhelmed. This approach supports engagement while minimizing practitioner workload.

[In MoveHealth]...**participants [can] review key outcomes, compare against Norms and track changes** over time without being overwhelmed.



Practitioner testing a participant and showing results in both the ForceDecks and MoveHealth apps.

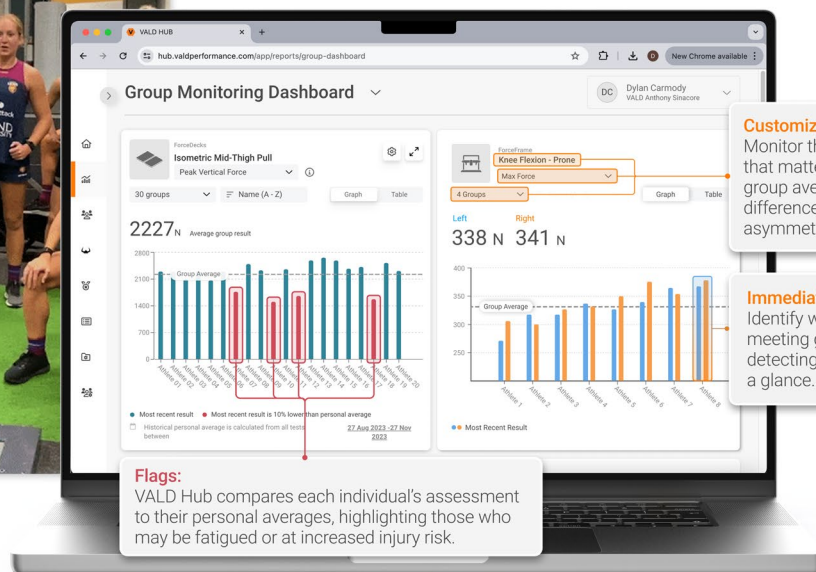
In other settings, practitioners may choose alternative delivery methods. Individual or group reports can be generated directly from platforms like VALD Hub as print-ready or PDF summaries or incorporated into custom reports using VALD's APIs for streamlined data transfer.

Together, these options enable practitioners to align reporting methods with their environment, maintaining consistency, security and efficiency.

Coach and Team Summaries

Practitioners can generate automated squad-level summaries directly within VALD Hub using the [Leaderboard](#) and [Group Dashboard](#), highlighting key metrics such as group averages and top performances.

These summaries can be shared as PDF reports with coaching or medical staff or accessed through various login permissions within VALD Hub, providing a clear and efficient overview of group performance without additional data handling.



Customizable Dashboard: Monitor the groups, tests and metrics that matter to you, while tracking group averages and individual differences in force production, asymmetry and more.

Immediate Comparisons: Identify who is above, below and meeting group averages, all while detecting new asymmetry changes at a glance.

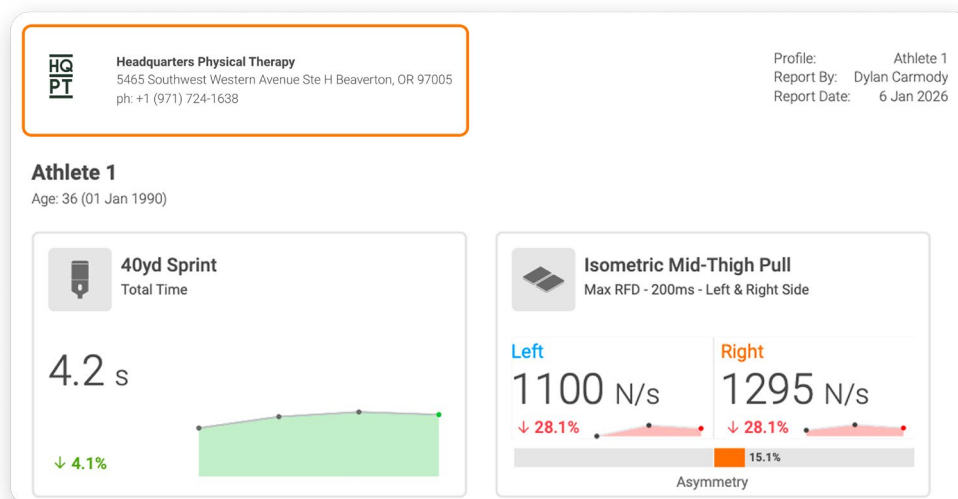
Flags: VALD Hub compares each individual's assessment to their personal averages, highlighting those who may be fatigued or at increased injury risk.

Stakeholder Communication

Large organizations, combine operators and clinics, often look for high-efficiency result-sharing options to reduce reporting burden. Cloud-based reporting platforms, such as VALD Hub, enable simple data-sharing options in both group summary and individual formats. PDFs and printable reports enable information sharing with scouts, medical teams and other stakeholders without excessive time burden.

Organization logos are automatically included, reinforcing brand identity while delivering clear, professional outputs across sessions and events.

Organization logos are automatically included [in participant profile PDFs from VALD Hub], **reinforcing brand identity while delivering clear, professional outputs across sessions and events.**



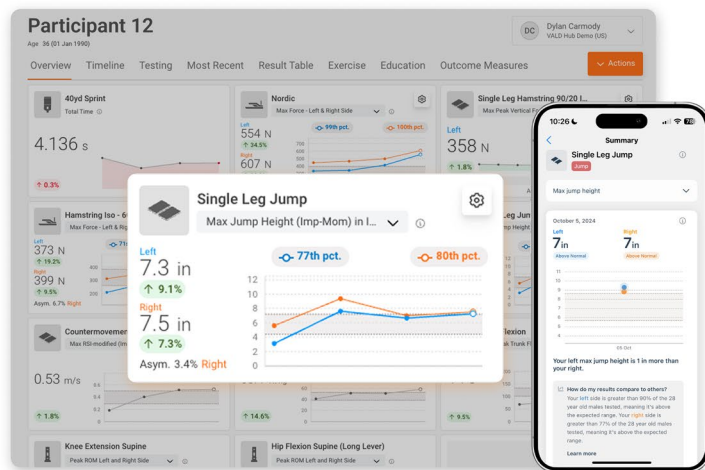
Organization logo printed on all VALD Hub reports.

Normative Data

Some databases, such as VALD's Norms, will populate immediately after a test is executed (e.g., CMJ testing on ForceDecks), providing real-time normative data comparisons for participants. These normative data analyses can also be simplified and automatically paired with participant-friendly applications like MoveHealth to provide simple explanations for each participant's results.

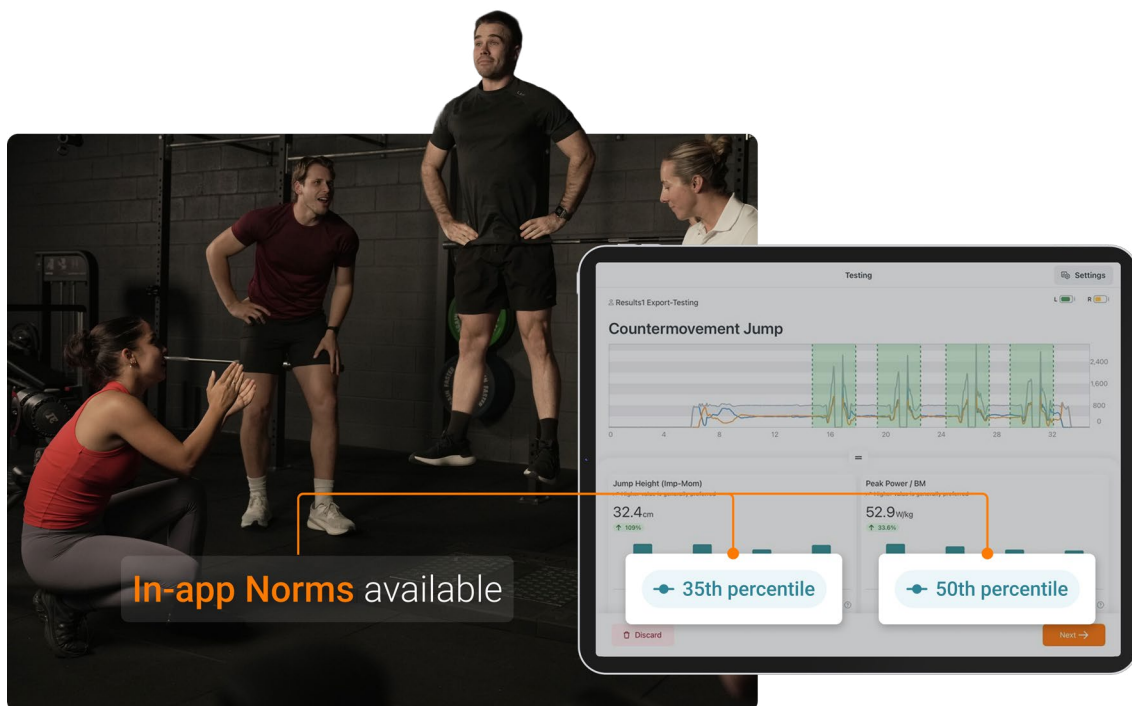
These normative data analyses can also be simplified and **automatically paired with participant-friendly applications like MoveHealth...**

For practitioners using printable or PDF reports to send to stakeholders, digital ecosystems like VALD Hub house both dashboard-visualized Norms and Norm comparisons in printed reports, allowing stakeholders to better understand how each participant ranks among their peers.



For practitioners seeking deeper context, [VALD's Normative Data Reports](#) support more advanced comparison and storytelling. These population-specific reports can be used alongside exported testing data to compare groups against higher-level benchmarks, such as aligning combine results with professional league datasets.

[Normative Data Reports] can be used **alongside exported testing data to compare groups against higher-level benchmarks...**

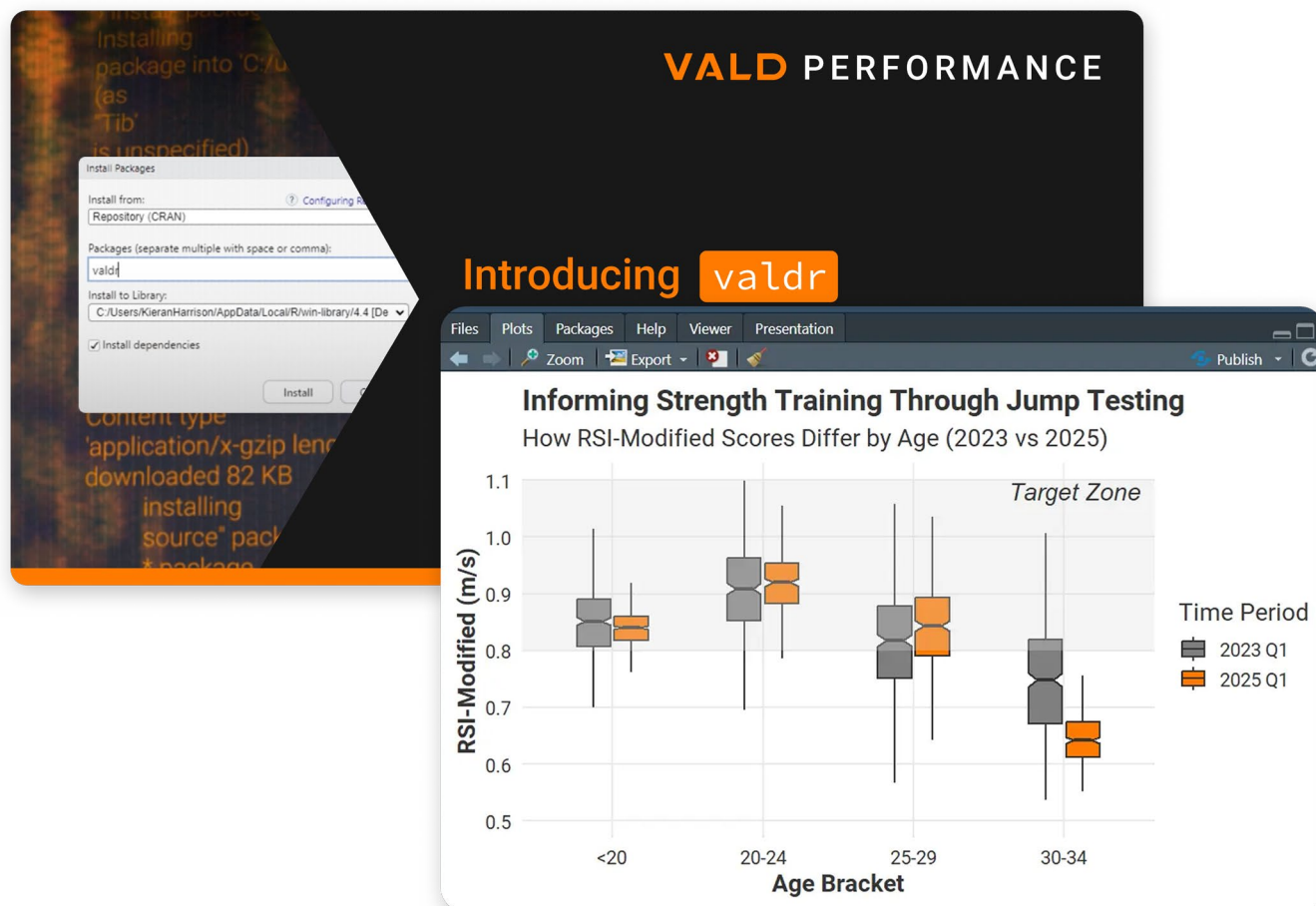


For practitioners interested in comparing their participant data with Normative Data Reports, external tools, such as API-driven workflows or [valdr](#), are required. Exporting data into analytics platforms enables practitioners to run personalized testing to compare participant data against any uploaded dataset.

Advanced Analytics

For researchers and analysts, open-source R packages such as [valdr](#) provide programmatic access to VALD Hub data. These platforms enable advanced statistical analysis, modeling and custom dashboards built entirely outside the platform.

For example, a sport scientist may use valdr to examine relationships between CMJ metrics and sprint performance across a season, with analyses updating automatically as new data is added to VALD Hub.



Choosing the **Right Reporting Path**

Choosing the right reporting path depends on how data is consolidated and who needs to use it. In many settings, testing data is collected across multiple systems, requiring a clear plan for whether results are reviewed in a central platform, integrated into an existing management system or exported for further analysis. Some practitioners prioritize quick, automated summaries, while others rely on tools such as Excel, Power BI or custom dashboards for more detailed exploration.

As described in the [Practitioner's Advanced Guide to Force Plates](#), group testing users commonly fall into one of three categories. These categories can help users streamline their reporting and data analysis workflows.



At VALD, we want to enable users to work with their data in ways that best suit them, both now and as their skills evolve. We broadly categorize **users' preferences into three personas.**

Matthew Roodvelt | Chief Product Officer, VALD

1
CATEGORY

"Do it for me"
Many of our users are short on time and need their data served up in a simple, easy-to-understand format.

FORCEDECKS
DUAL FORCE PLATE SYSTEM
movehealth+

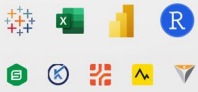
2
CATEGORY

"Help me do it"
Other users want more than the defaults, so we offer intuitive customizations into our reporting, allowing them to modify what they see without requiring any special technical skills.

VALDHUB
NORMS

3
CATEGORY

"Let me do it"
Some of our users are highly technical and prefer to use third-party data analysis platforms such as Excel, R Studio, Power BI or Tableau to perform their analyses.



Reporting should also match the audience. Participants may benefit from simplified [results], while...stakeholders often require [deeper analysis]...

Reporting should also match the audience. Participants may benefit from simplified, app-based results (e.g., [MoveHealth](#)), while coaches, practitioners and stakeholders often require group summaries, benchmarks and trends. More advanced users may extend analysis through exports or integrations that combine testing data with other performance or health information.

Case Studies

EMP Onsite: **Workplace Wellness Screening**



Dale Turner

Managing Director

EMP Onsite

EMP Onsite is a workplace health provider operating primarily in manual and high-risk industries. Their mission is to move beyond reactive injury management and instead identify leading indicators of injury risk before symptoms emerge. By combining objective physical capacity testing with task-based risk analysis, EMP aims to prevent injury, reduce lost work time and provide measurable value to employers at scale.

Common Challenges in the Workforce

The populations EMP supports commonly experience productivity and cost challenges driven by upper-limb and repetitive manual task injuries. Risk is often concentrated in new starters, workers with prior injury and aging employees, many of whom return to work with reduced capacity.

Recognizing that traditional programs are largely reactive, EMP identified the need for earlier, objective insight into injury risk to better guide intervention and measure outcomes.

Assessment Strategies

EMP's screening model integrates task exposure data with objective physical capacity testing.

Portable and scalable measurement technologies, such as dynamometry and isometric strength testing, are used to assess joint- and muscle-specific capacity.

Grip, shoulder strength and IMTP data are collected at scale, benchmarked against normative values and tracked over time to identify risk patterns. Assessments are delivered by a multidisciplinary team using consistent, objective criteria.



Key Findings and Interventions

Screening consistently identified three high-risk groups. With these populations identified, EMP began testing. Objective data enabled EMP to group workers based on deviations from role- and age-specific benchmarks, rather than relying on symptoms alone.

Interventions were then matched to risk level:

- **Lower to Moderate Risk:** Workers were managed through group-based strengthening and conditioning programs, improving efficiency and scalability.
- **High Risk:** Workers received targeted, individualized intervention informed by in-depth assessment of contributing factors such as joint strength, posture and movement capacity.

Objective retesting allowed EMP to quantify response to intervention, with average functional improvements of approximately 22% across participants, supporting confident progression back toward role-specific benchmarks.

Objective retesting allowed EMP to quantify response to intervention, with **average functional improvements of approximately 22% across participants...**

Return on Investment

Objective measurement significantly improved EMP's ability to demonstrate return on investment (ROI). In one high-risk port and shipping workforce, injury claims fell from 14 per month to zero over six months, accompanied by a reduction in workers' compensation costs of more than \$30,000 per month.

Free eBook



Practitioner's Guide to ROI

A comprehensive guide for practitioners looking to maximize their ROI from objective measurement technology.

VALD

Practitioner's Guide to **ROI**

A comprehensive guide for practitioners looking to maximize their ROI from objective measurement technology.



When modeled over time, this equated to an estimated 5.5:1 ROI (approximately \$900,000 in value over six months) through reduced claims and premium impact. Internally, the same data improved staff confidence and credibility, enabling clearer, evidence-based communication with executives and boards.

When modeled over time, this equated to an estimated **5.5:1 ROI (approximately \$900,000 in value over six months)** through reduced claims and premium impact.

Outcome and **Practical Takeaways**

By embedding objective measurement into workplace health screening, EMP shifted from assumption-based decision-making to data-driven confidence. Large-scale physical testing using [VALD technologies](#) allowed early risk identification, precise intervention targeting and confident outcome quantification.

Key takeaways for practitioners and organizations:

- Objective screening enables proactive injury prevention rather than reactive care.
- Normative benchmarking strengthens risk stratification across diverse workforces.
- Scalable testing supports both individual and group-level intervention strategies.
- Clear, outcome-driven reporting underpins strong ROI conversations with stakeholders.

Private Combine Organization: **Thrive Kelowna**



Dr. Samuel Spinelli

High Performance Coordinator

Thrive Kelowna

In 2024, [Thrive Kelowna](#) delivered a large-scale basketball combine designed to replicate the structure and standards of collegiate and professional testing environments.



The primary objectives were to expose participants to next-level testing protocols, foster meaningful engagement and establish an objective baseline to support long-term development.

Learn more about Thrive Kelowna's testing and combine planning process [here](#).

The Challenge of High-Volume Testing

High-volume combines place significant demands on organization, consistency and data quality. Thrive addressed this challenge by prioritizing preparation and partnership. Collaborating with a local club program, **Goodwin Hoops**, allowed Thrive to focus on testing and performance delivery while the club supported recruitment and scheduling, ensuring alignment and steady participant flow.

Preparation and Workflow Design

Preparation began before the event day. Rosters were collected in advance and preloaded into VALD Hub, with each participant assigned a unique identifier. On the floor, participants were referenced by number rather than name, allowing staff to locate profiles instantly and avoid station delays. This simple workflow decision significantly reduced bottlenecks and maintained testing tempo.

Staff were briefed on standardized protocols, station flow and communication strategies to ensure consistent execution with high testing volumes.

Staff were briefed on standardized protocols, station flow and communication strategies to **ensure consistent execution with high testing volumes.**



Test Selection and Station Flow

The testing battery was intentionally streamlined to prioritize quality over quantity. After a standardized warm-up, participants rotated through stations that assessed jump performance, hop capacity, anthropometrics, linear speed and change of direction (COD). Jump and hop testing were captured using [ForceDecks](#), while linear speed and agility were assessed with [SmartSpeed](#).



Participants completed a fixed number of attempts at each station before rotating, minimizing downtime and decision fatigue. Groups progressed sequentially through the testing area using a staggered distribution, supporting participant experience and data integrity.

Turning Data into Insight

Since data were captured digitally and centrally, Thrive could move directly from testing to interpretation. Results were contextualized using normative benchmarks to answer practical questions around performance level, allowing participants to be categorized based on [VALD Norms](#) to improve training and rehabilitation decisions.

Summaries were shared with participants, coaches and parents, reinforcing transparency and long-term engagement.

Practical Takeaways

Successful group testing is built on a small number of repeatable principles that determine whether data collection translates into meaningful insight:

- Advance preparation, including roster preloading and defined station flow, reduces bottlenecks and maintains testing tempo.
- Intentional test selection ensures assessments reflect sport-specific qualities without unnecessary complexity.
- Standardized protocols and staff communication preserve data quality as testing volume increases.
- Centralized data capture and reporting enable results to be contextualized and shared efficiently after the event.

When these elements are aligned, high-volume testing becomes a scalable process for insight, engagement and long-term athletic development rather than a one-off data collection exercise.

...high-volume testing becomes a scalable process for insight, engagement and long-term athletic development rather than a one-off data collection exercise.

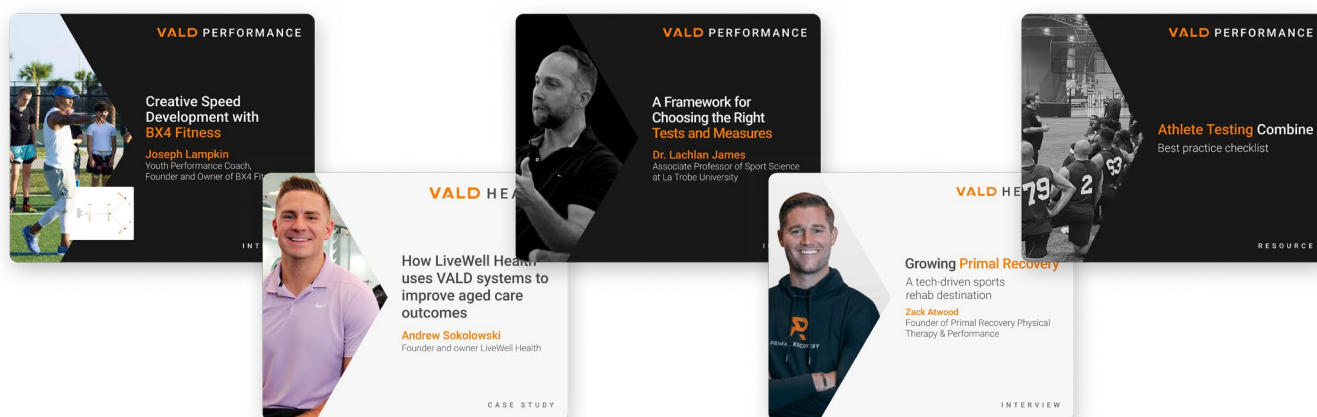
What Next?

Group testing has evolved into a repeatable, high-impact workflow when supported by the right planning and resources. Practitioners looking to implement or refine group testing can draw on established frameworks and [VALD Resources](#), such as the [VALD's Athlete Testing Combine Best Practice Checklist](#), [Practitioner's Series](#) and published [VALD articles](#).



Similarly, learning from organizations and practitioners who regularly run combines, including programs such as [Elite11](#), [Prep Baseball](#) and [Thrive Kelowna](#), can also help inform structure, test selection and event execution. When preparation, communication and technology are aligned, group testing enables practitioners to scale assessments without compromising data quality or participant experience.

With the right references and systems in place, group testing moves beyond logistics and becomes a reliable tool for insight, engagement and long-term performance and health decision-making.



Additional Resources

The following resources provide practical tools to support efficient, standardized and scalable group testing. Each can be adapted to the specific needs of your environment – whether you are working with a small academy squad, a professional team or a large-scale combine.

Sample Schedules and Checklists

Event Day Timeline – Examples

Equal Distribution Approach				
Time	Group A	Group B	Group C	Group D
08:00-08:15	Group warm-up	Group warm-up	Group warm-up	Group warm-up
08:15-08:30	Station 1: Range of motion (ROM)	Station 2: IMTP and CMJ	Station 3: $\frac{3}{4}$ -court sprint	Station 4: Nordic hamstring curl
08:30-08:45	Station 2: IMTP and CMJ	Station 3: $\frac{3}{4}$ -court sprint	Station 4: Nordic hamstring curl	Station 1: ROM
08:45-09:00	Station 3: $\frac{3}{4}$ -court sprint	Station 4: Nordic hamstring curl	Station 1: ROM	Station 2: IMTP and CMJ
09:00-09:15	Station 4: Nordic hamstring curl	Station 1: ROM	Station 2: IMTP and CMJ	Station 3: $\frac{3}{4}$ -court sprint
09:15-09:30	Buffer/exit	Buffer/exit	Buffer/exit	Buffer/exit

Staggered Distribution Approach				
Time	Group A	Group B	Group C	Group D
08:00-08:15	Warm-up	–	–	–
08:15-08:30	Station 1: ROM	Warm-up	–	–
08:30-08:45	Station 2: IMTP and CMJ	Station 1: ROM	Warm-up	–
08:45-09:00	Station 3: ¾-court sprint	Station 2: IMTP and CMJ	Station 1: ROM	Warm-up
09:00-09:15	Station 4: Nordic hamstring curl	Station 3: ¾-court sprint	Station 2: IMTP and CMJ	Station 1: ROM
09:15-09:30	Exit	Station 4: Nordic hamstring curl	Station 3: ¾-court sprint	Station 2: IMTP and CMJ
09:30-09:45	–	Exit	Station 4: Nordic hamstring curl	Station 3: ¾-court sprint
09:45-10:00	–	–	Exit	Station 4: Nordic hamstring curl
10:00-10:15	–	–	–	Exit

Equipment Checklist

- ☒ ForceDecks Lite or Max units (level checked and fully charged)
- ☒ SmartSpeed Plus gates and tripods (verified alignment)
- ☒ ForceFrame attachments matched to planned tests
- ☒ SmartJump mats or DynaMo devices for overflow stations
- ☒ Extension leads, charging cables and spare batteries
- ☒ Laptop or tablet logged into VALD Hub (pre-configured test batteries)
- ☒ Tripod labels and station signage for clarity

Staffing Checklist

Role	Responsibility
Lead Tester	Oversees testing order, flow and quality control
Tech Manager	Performs device setup, calibration and troubleshooting
Participant Coordinator	Manages registration and rotation timing
Data Monitor	Reviews real-time data in VALD Hub
Support Staff	Supervises warm-up, assigns a testing station and provides participant feedback

Communication Templates

Pre-Event Email to Participants

Hi team,

We'll be running our next testing session on **[date] at [location]**. Please arrive 10-15 minutes early to allow for check-in and warm-up.

What to bring: Training kit, running shoes and water bottle.

What to expect: You'll complete a series of tests, including [insert tests here]. Each station will be explained beforehand.

After testing: Your individual results will be available through the MoveHealth app – keep an eye on your inbox for an invite to set up your account.

Thanks for your effort and professionalism. We look forward to seeing you there.

– [Your Name]

Department | Organization

Pre-Event **Staffing Brief**

Before the event, testing organizers should ensure all staff understand the following:

- ✓ Session objectives and testing order review
- ✓ Roles and station assignments confirmation
- ✓ VALD Hub login and test battery verification
- ✓ Safety and participant flow reminders
- ✓ Contingency plan (if technology issues occur)

Test and Metric Templates

Station	Assessment Focus	Technology	Common Metrics Captured
Station 1	Squat, lunge, broad jump	HumanTrak	Joint ROM, segment angles, movement deviations, jump distance
Station 2	Hip Add/Abd, shoulder external and internal rotation (ER/IR)	ForceFrame	Peak force, force relative to body mass, left-right asymmetry
Station 3	CMJ, hop test	ForceDecks	Jump height, peak power / body mass, reactive strength index-modified (RSI-Mod), reactive strength index (RSI)
Station 4	30m sprint	SmartSpeed	30m time; 10m and 20m splits
Station 5	Nordic hamstring curl	NordBord	Peak force, left-right asymmetry, impulse (optional)

Further Reading and Support

VALD Knowledge Base

Access step-by-step guides and technical support for all VALD systems:

<https://support.valdperformance.com>

VALD Academy

Online learning platform with courses covering all VALD systems, masterclass videos and more:

<https://resources.vald.com/vald-academy>

Notes

Group testing has **a lot going on.**

Testing large groups introduces logistical, technical and communication challenges – from managing flow and fatigue to maintaining data quality and delivering meaningful outputs at scale.

Drawing on real-world examples across sport, health, workplace and education settings, this guide distills best practice for planning, executing and reporting on group testing sessions using objective measurement technology.

Some questions we'll answer in this guide:

Q: How do I plan an **effective group testing event**?

Q: Which testing options are best suited for **high-volume environments**?

Q: How do I manage participant flow and **logistics at scale**?

Q: How can I interpret, report and deliver group testing data effectively?



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Want to learn more? We are here to help.

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